

Red Tide — Arcade Game 5 Design Document

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Overview

Red Tide is a four-player 3D survival shooter in the Deep Arcade, built on the bones of Black Ops 2 and 3 Zombies but with no zombies: the players are divers dropped into living ecosystems, earning money by killing beasts while the food web, the blood in the water, and the enemies drawn by their gunfire close in around them. Every map is an ecosystem of about 40 beasts and 12 flora that eat, hide, breed, and hunt each other whether or not the players are watching.

What it keeps from Zombies

Zombies system	Red Tide version
Rounds	Tides. Each tide raises the ecosystem's aggression, the enemy pressure, and the bounties. A tide ends when its kill quota is met
Points	Scrip , earned per hit and kill, spent on doors, racks, the Locker, tonics, and the Forge
Wall buys	Racks: weapons chalked on the walls of the map
Mystery Box	Davy's Locker: a sea chest that deals a random weapon and moves when the eel inside gets bored
Pack-a-Punch	The Pressure Forge: upgrades weapons and adds alternate ammunition
Perks (Juggernog, Speed Cola, Quick Revive, Double Tap, Stamin-Up, PhD, Deadshot, Mule Kick, Electric Cherry, Widow's Wine, Vulture Aid, Tombstone, Who's Who)	Tonics: bottled brines from vending machines, each a reimagining of one perk
Gobblegums	Salt Charms: single-use charms earned between matches

Zombies system	Red Tide version
Power-ups	Drops: Resupply, Blood Frenzy, Double Scrip, Purge, Shipwright, Fire Sale, Harpoon Hour
Buildables	Salvage builds: shield, turbine, tripwire nets, decoy buoys
Downs, revives, Last Stand pistol	Unchanged, underwater
Special rounds (hellhounds)	Hunts: an enemy faction round, or a predator round when the water is bloody enough
Easter eggs and lore	Dossiers and one hidden quest per map

Design pillars

1. **The food web is the enemy.** Zombies come because it's a round; beasts come because you made blood, noise, or a corpse. The ecosystem runs without you and reacts to everything you do.
2. **Every kill is a decision.** Small kills are safe scrip; big kills bring bigger things; too many kills bring the things that eat everything.
3. **Enemies are prey too.** The human and unhuman factions that hunt you are also on the menu, and the smartest play is often to let the reef eat them.
4. **Each map is an ecosystem first and a level second.** Layouts are built around where creatures live, feed, and hide.
5. **Zombies pacing.** Fast, readable, arcade-loud: 60-second tides early, minutes-long late, always a way to run, always a reason to stop and buy.

The five maps (smallest to largest): the Sunken Ship, the Underwater Cave, the Coral Reef, Atlantis the Forgotten City, and Approaching the Void.

Session: 4 players (or 1–3 with scaled quotas), 20–60 minutes, ending when every player is dead. The score is the tide reached and the scrip earned, and both pay arcade tokens.

Core loop and rules

A match runs in tides. Each tide has a kill quota; meeting it ends the tide, pays a bonus, and starts the next one harder. Between tides the water calms for 20 seconds, the only reliable time to buy, build, and revive.

The spawn-in (identical every match)

Slot	Item
Sidearm	The Cormorant, a brass gas-pistol: 8 darts, 32 reserve
Melee	Diver's knife
Tactical	2 limpet charges (stick to surfaces or beasts, 3 s fuse)
Scrip	500
Lamp	Helmet lamp, switched on
Build slot	Empty

All four divers spawn in the map's start pocket, a small safe room with one rack pistol, one door out, and the first tonic machine visible through the door.

Scrip

Action	Scrip
Hit a beast or enemy	10
Kill	The beast's bounty (40–1,500, in each map's table)
Headshot or weak-point kill	Bounty +50%
Melee kill	Bounty +30
Repair a barricade net	10 per plank, max 100 per tide
Revive a teammate	100
Tide cleared	100 × tide number, split among the living
Enemy faction kill	Bounty 200–600 plus their dropped weapon

Bounties rise 10% per tide. Scrip is per player and can't be traded or dropped; the only way to even it out is the Fair Shares charm.

Tides

Tide	Kill quota (4 players)	What changes
1-3	12 / 16 / 22	Ecosystem calm; small and mid beasts only; no enemies
4-9	28-60	Predators respond to blood; enemy attraction on; a Hunt every 5th tide
10-19	70-150	Apex beasts wander; enemy squads carry upgraded weapons; blood decays slower
20+	+20 per tide	Everything at once; Blood Frenzy needed to keep pace; the map's boss beast can appear at any time

Quota scales 0.4 / 0.6 / 0.8 / 1.0 with 1-4 players. Kills by beasts eating enemies or each other do not count; kills by players' builds do.

Health, downs, revives

- 100 HP, regenerating 4 s after the last hit at 20 HP/s (Zombies-style). Juggernaut Tonic raises HP to 250.
- At 0 HP a diver is **downed**: they float with the Cormorant only, for 45 s, bleeding into the water (a downed diver is a strong blood source, which is the danger and the drama).
- Any teammate can **revive** in 4 s (2 s with Quick Brine). A revived diver loses all tonics unless they hold a Salt Charm that keeps them.
- If every diver is down, the match ends. Solo has one self-revive per Quick Brine bought (three per match).
- **Air**. There is no air timer; divers breathe indefinitely. The pressure against camping in a corner comes from blood, scent, and the enemy alarm instead.

Doors, power, and the map opening up

- **Doors and debris** cost 750-2,500 scrip and open permanently for everyone. Each map opens in stages from the start pocket outward.
- **Power** (the generator, the tide mill, the temple's crystal) turns on the Forge, the Locker's second location, tonic machines beyond the first two, and the map's traps. Turning on power also wakes the ecosystem's larger predators, so the choice of when

to do it is real.

- **Traps** cost 1,000 per use and kill by the map's rules: a cargo crane, a stalactite drop, a coral gate, a flooding chamber, a lab's pressure door.

Drops

Some beasts drop a glowing bottle on death (about 1 in 40 kills, weighted away from repeats).

Drop	Effect
Resupply	Full ammo for everyone
Blood Frenzy	30 s: every hit kills; the water fills with blood at 5× rate (huge scrip, huge danger)
Double Scrip	30 s
Purge	Every beast within 40 m dies and leaves no blood; enemies flee
Shipwright	All barricade nets repaired
Fire Sale	30 s: the Locker costs 10 and appears at every location
Harpoon Hour	30 s: the Harpoon Cannon in hand, infinite bolts

Hunts

Every 5th tide from tide 5 is a **Hunt**: instead of a quota, the map's enemy faction arrives in force (see Enemy factions) and the tide ends when they're dead or eaten. Starting at tide 10, a Hunt can also be a **Predator Hunt**: the map's apex beasts come to the players' position because the water is bloody enough, and the quota is a fixed number of apex kills. The reward for either is a Resupply and a guaranteed Locker weapon.

The ecosystem engine

Every map runs one simulation, **Ecosystem**, in which about 40 species and 12 flora live by a shared set of rules: what they eat, what eats them, what they smell, what they hear, where they live, and who they help. Players and enemy factions are members of the same

web. The engine is built so that a map's species list is data (`redtide_species.json`) and the rules are the same everywhere.

A species record

Field	Meaning
Class	Fish, reptile, crustacean, amphibian, mammal, insect, cephalopod, cnidarian, echinoderm, worm, bird, enemy
Size	1 (finger) to 6 (leviathan); a predator eats prey up to 60% of its size, or larger when it hunts in a school or pack
Trophic tier	0 flora, 1 grazer or filter feeder, 2 small predator, 3 mid predator, 4 apex, 5 boss; scavengers and parasites carry a tag
Diet	Species and tags it eats, each with a preference weight
Blood yield	Blood emitted per second when wounded ($= \text{size} \times \text{wound fraction} \times 2$) and on death ($= \text{size} \times 20$, then a trickle from the corpse)
Blood threshold	How much blood in the water makes it investigate (small scavengers 5, mid predators 30, apex 120, the boss 400)
Senses	Sight range, scent sensitivity, sound sensitivity, and whether it has electroreception (sees players through walls at short range)
Temperament	Aggression, fear, curiosity (0–1 each)
Home	Flora or terrain it lives in, nest radius, whether it defends its home
Social	Solo, pair, school (boids), pack (coordinated), swarm
Bounty	Scrip on kill
Drops	Chance of a drop bottle and any special part

The seven layers of the web

1. **Trophic layer.** The diet graph. Each beast has a hunger meter; hunger raises the weight of hunting over everything else. Grazers eat flora; predators eat by preference and size; scavengers eat corpses. A well-fed predator ignores prey, so an area that has just been hunted through is briefly safer.
2. **Blood and scent layer.** The map holds a 3D scent grid (2 m cells). Blood diffuses to neighboring cells each second and decays 2% per second (1% at tide 10+, 0.5% at tide 20+, so late tides never clear). Every species samples the grid within its scent range each second; when the total exceeds its threshold, it follows the gradient to the source. Because thresholds are per species, one dead minnow brings scavengers, a dead tuna brings sharks, and four dead tuna at once brings the thing that eats sharks.
3. **Sound layer.** Every weapon has a noise value; explosions and traps are louder. Sound is a second field with fast decay (20% per second). Species react by temperament: fearful ones flee it, curious ones approach it, aggressive ones attack it. Sound also feeds the enemy alarm (below).
4. **Territory layer.** Beasts with a home defend a radius around it against anything larger than prey, including players and enemies, and return home when calm. Nests, burrows, anemone beds, and wrecks are where fights start without any blood at all.
5. **Symbiosis layer.** Cleaners reduce the aggression of hosts they service; hosts shelter symbionts; parasites attach to beasts and divers (slowing, bleeding, or blinding them until scraped off at a cleaning station). Mimics look like harmless species until touched. Lures and lights draw curious species.
6. **Flora layer.** Flora grows, gets grazed, shelters, feeds, and sometimes hunts. Flora density sets grazer carrying capacity; grazer numbers set predator patrol frequency; so shooting up a kelp bed or a coral head changes what lives there for the rest of the match.
7. **Enemy layer.** Enemy factions are species in the same web: they bleed, they are prey to apex beasts, they hunt beasts that threaten them, and some of them feed the web on purpose (chumming, netting).

The blood rules the player learns

- A shot beast bleeds; a dead one bleeds much more. Blood attracts by total amount in the area, not by the number of kills, so ten small fish and one big fish can call the same predator.
- Corpses keep bleeding until scavengers finish them (30–120 s by size). Scavengers, therefore, are the players' friends, and killing them for scrip is the classic mistake.

- Wounded players bleed; downed players bleed heavily. Reviving fast is also ecology.
- Moving out of bloody water is the only real reset; blood follows currents, so the map's flow tells you where it goes.
- The HUD shows blood as a red tint at the edge of the mask and a small scent meter, and predators' approach as a pulse from their direction.

The enemy alarm

- Each map region has an **alarm** value. Gunfire adds the weapon's noise; explosions add triple; melee adds nothing. It decays 5% per second.
- When alarm exceeds $100 \times (1 / (1 + 0.1 \times \text{tide}))$, a roll every 5 s spawns an enemy patrol at the region's faction entry with chance $10\% + 3\% \times \text{tide}$ (capped at 70%). Higher tides make quieter play matter more.
- Four divers firing in one region fill one alarm four times faster than four divers spread across four regions. Spreading out lowers enemy pressure but raises blood-in-one-place risk when someone goes down alone: the coordination trade-off the Sunken Ship teaches first.
- Enemies that arrive follow the same sound and blood senses as beasts, so they find you the way the beasts do.

Behavior state machine

Every beast runs: **Graze or Rest** → **Investigate** → **Hunt** → **Feed** → **Flee** → **Defend** → **Return** → **Dead**, with transitions driven by hunger, blood, sound, threat size, and home distance. Schools use boids with a shared leader; packs assign flank and drive roles; swarms are a single agent with many bodies. Fleeing beasts drop the blood they were about to make: a wounded fish that escapes leaves a thin trail, which is itself a lure for the next predator.

Population

- Each region has spawn pools per species with a carrying capacity tied to flora density and tide number.
- Beasts that leave a region to follow blood are replaced slowly at home; beasts killed in a region are replaced at the region's respawn rate (small fast, apex slow, boss never).
- Predators that overfeed leave; grazers then rebound; predators return. The cycle is tuned to about 3 minutes so a match feels alive without stalling quotas.
- The **boss beast** of each map is a single persistent agent with a very high blood threshold and a home; it wanders more each tide.

Simulation budget

About 300 active agents at full rate (10 Hz decisions, 60 Hz movement) near the players, and population counters only for regions beyond 120 m. The host runs the simulation; clients receive agent positions and states with fog filtering by distance. `--eco-sim <map> <minutes>` runs the ecosystem headless and prints population curves, blood levels, and predator arrival times for a scripted player pattern.

Weapons

Red Tide's arsenal follows Zombies' shape (a fixed starting pistol, wall racks, a random box, a Pack-a-Punch, and a wonder weapon per map) with underwater guns: gas-driven dart and needle guns, spearguns, harpoon rifles, and sealed-cartridge powder guns. Every weapon has a noise value that feeds the alarm and the sound layer, so the loud guns are the strong ones and the quiet ones are the safe ones.

Stats and rules

- **Two weapon slots** (three with Mule Brine). Weapons are picked up from racks, the Locker, or dead enemies.
- **Damage** is per dart or bolt; **rate** is rounds per minute; **noise** 1–10 feeds the alarm; **reload** is seconds.
- Ammo is bought at the rack (half the weapon's price) or refilled by a Resupply drop. Locker weapons can only be refilled by Resupply or the Forge's re-arm.
- **Headshots and weak points** do 2× (3× with Deadeye Brine).
- **Water drag**: dart weapons lose damage past 25 m, harpoons past 60 m, powder guns past 40 m.

Racks (wall weapons)

Weapon	Type	Damage	Rate	Mag / reserve	Noise	Reload	Scrip	Notes
The Cormorant	Gas pistol (start)	30	300	8 / 32	2	1.4	—	The Last Stand pistol; Forged it becomes the Albatross, 3-dart burst
Gannet	Gas pistol	40	280	10 / 50	2	1.5	500	Rack in every start pocket

Weapon	Type	Damage	Rate	Mag / reserve	Noise	Reload	Scrip	Notes
Needler Mk I	Needle repeater (SMG)	22	900	30 / 150	4	2.0	1,000	Early workhorse
Sea-Pattern Carbine	Powder carbine	45	600	24 / 168	6	2.2	1,200	Loud, accurate
Flechette 12	Scatter gun	12 × 8	80	6 / 36	7	3.0 (shell by shell)	1,200	Close corridors
Long Speargun	Speargun (marksman)	120	60	1 / 20	3	1.8	1,500	Quiet one-shot on small fish
Needler Mk II	Needle repeater	26	950	40 / 200	4	1.9	1,500	
Trawlerman's Rifle	Powder rifle	70	420	20 / 140	6	2.4	1,750	
Bolt Harpoon	Harpoon rifle (sniper)	260	40	1 / 16	5	2.5	2,000	Pins small beasts to walls; retrievable bolts
Chumthrower	Support	0	—	10 / 30	1	2.0	1,500	Fires chum: creates blood on a spot to steer predators, a deliberate lure
Boarding Axe (melee rack)	Melee	180	—	—	0	—	2,500	Replaces the knife

Davy's Locker (the box)

950 scrip a pull; a moray eel inside shuffles the guns and, after 8–12 pulls, snaps the chest shut and it sinks to another location (a lantern buoy marks the new one). Fire Sale puts it everywhere for 10. The Locker deals from this pool plus every rack weapon and the map's wonder weapon (2% per pull, guaranteed within 12 pulls if not yet held by the team).

Weapon	Type	Damage	Rate	Mag / reserve	Noise	Notes
Gatling Needler	Needle LMG	24	1,100	120 / 360	5	Spins up 0.5 s

Weapon	Type	Damage	Rate	Mag / reserve	Noise	Notes
Twin Gannets	Akimbo gas pistols	40	560	20 / 100	3	
Stormlock	Powder auto-rifle	55	720	30 / 210	7	
Drum Flechette	Auto scatter gun	11 × 8	180	12 / 48	8	
Reef Rattler	Burst needle rifle	35 × 3	3-burst	27 / 189	4	
Cannon Harpoon	Heavy harpoon	400	30	1 / 12	8	Explodes on impact with a beast over size 3
Limpet Launcher	Launcher	350 splash	40	4 / 12	9	Charges stick and blow
Net Gun	Utility	0	60	4 / 12	2	Nets hold a beast up to size 3 for 6 s (size 4 for 3 s)
Tesla Gaff (melee)	Melee	250 + stun	—	—	3	Chains to 2 nearby
Trident (melee)	Melee	300	—	—	0	Reach; lunge on sprint

Wonder weapons (one per map, Locker only)

Map	Weapon	What it does
The Sunken Ship	The Galvanic Rod	A ship's dynamo on a pole: an arc that chains through everything in 10 m, stunning; Forged: the Lightning Keel, arcs leave a 5 s electric field
The Underwater Cave	The Resonator	A sonic cannon: a cone that kills small beasts outright and knocks large ones back, and shatters stalactites onto enemies; Forged: the Deep Bell, its ring also cancels blood scent in the cone for 5 s
The Coral Reef	The Anemone Gun	Fires living polyps that root where they land and grab anything that passes; Forged: the Garden, polyps also poison and spread once

Map	Weapon	What it does
Atlantis	The Tide Staff	Bends the current: a wave that carries beasts and enemies away, or pulls them into a trap; Forged: the Sovereign's Staff, the wave drowns enemies in air pockets and carries blood with it
Approaching the Void	The Abyssal Lure	A lantern that draws every beast in 60 m to a point for 8 s, then bursts; Forged: the Deep Call, the burst also calls one apex beast to the point on your side for 30 s

The Pressure Forge (Pack-a-Punch)

5,000 scrip after power. A pressure chamber that crushes the weapon into a stronger form: +150% damage, +50% ammo, a new name, a sea-glass finish, and a slot for one **alternate ammunition** (rolled free, re-rolled for 2,500):

Ammunition	Effect (Zombies analogue)
Galvanic	Every 6th hit chains lightning to 3 nearby (Dead Wire)
Chum Rounds	Hits mark the target so predators prefer it for 10 s: the beast becomes bait (Turned)
Flare Burst	Hits explode in light, blinding beasts with eyes for 3 s (Fireworks)
Riptide	Every 6th hit pushes everything in a line 8 m (Thunder Wall)
Boiling	Hits cook: burn damage that leaves no blood (Blast Furnace)

The Forge can also **re-arm** a Locker weapon (full ammo) for 4,500.

Melee

Diver's knife (100 damage, silent), Boarding Axe, Tesla Gaff, Trident, and the Forge-only **Sawtooth** (a sawfish rostrum, 450, bleed). Melee kills make no alarm and less blood (the wound is one clean cut), so a knife team is the quiet route through the early tides.

Animations

- **Reloads** are underwater and slow-fast-slow: gas pistols pop the cylinder and thumb in darts one at a time until full or interrupted; needlers swap a glass cartridge with a hiss of bubbles; powder guns eject a sealed brass clip that sinks; spearguns cock with both arms and a visible strain; harpoon rifles winch a bolt from a side quiver; the scatter gun is loaded shell by shell and can be interrupted. Every reload has an empty and a partial version and a Speed Brine version 30% faster.
- **Firing:** bubble bursts from gas guns, a brass flash from powder guns, a line snapping taut from spearguns.
- **Inspect** animations show the sea-glass Forge finish.
- **Weapon skins:** arcade tokens buy finishes (barnacle, verdigris, bone inlay, pearl, red tide) that apply to all weapons.

The Harpoon Cannon (Harpoon Hour)

A shoulder-mounted salvage cannon with infinite bolts for 30 s, 500 damage, noise 10: the map's loudest sound, which is the joke and the risk.

Tonics, charms, items, builds, and progression

Tonics (perks)

Brass vending machines that dispense a bottle; four can be held (five with a charm). Each map has 6–8 of the 12 placed by its designer; the first two are always Juggernaut and Quick Brine.

Tonic	Zombies parent	Effect	Scrip
Juggernaut Brine	Juggernog	250 HP	2,500
Quick Brine	Quick Revive	Revives in 2 s; solo: self-revive (3 per match)	1,500 (solo 500)
Speed Brine	Speed Cola	Reloads and builds 30% faster	3,000
Double Shot	Double Tap	+33% rate of fire and darts count double	2,000

Tonic	Zombies parent	Effect	Scrip
Kick Brine	Stamin-Up	Faster swim, sprint longer, breaks free of grabs 50% faster	2,000
Deadeye Brine	Deadshot	Weak-point damage 3×, aim snaps to weak points	1,500
Mule Brine	Mule Kick	Third weapon slot	4,000
Bubbleskin	PhD Flopper	No self-damage; a dolphin dive from height bursts a blast	2,000
Shocking Salt	Electric Cherry	Reloading releases a shock that stuns beasts within 4 m	2,000
Widow's Ink	Widow's Wine	Melee and hits ink-cloud attackers; grenades become ink nets	4,000
Vulture's Eye	Vulture Aid	See scrip and ammo through walls; beasts sometimes drop ammo; scent meter shows predator direction earlier	3,000
Cold Coffin	Tombstone / Who's Who	On death, a shadow of you swims for 30 s and can revive your body with your gear	2,000

Salt Charms (gobblegums)

Earned between matches with arcade tokens and from tide milestones; a player brings 5 into a match and uses one at a time from a charm pouch.

Charm	Effect
Keep Your Brines	Tonics survive a down (once)
Salt Circle	20 s: beasts won't cross the player's position
Locker	Next pull is pressure forged: upgrades weapons and adds alternate ammunition to weapon picked up

Charm	Effect
Fair Shares	Every living diver's scrip is pooled and split evenly; the whole team sees the split (once per match)
Slick Fins	30 s of full sprint
Clean Water	Clears blood within 30 m
Fisher's Luck	Next 10 kills drop double scrip
Chum Bucket	Spawns a big chum cloud 30 m away: the reef looks over there
Hard Shell	60 s of 50% damage reduction
Ghost Fin	15 s: beasts and enemies lose the player's scent and sound

Items

Item	Slot	Use
Limpet charge	Tactical (2-4)	Sticks and blows; the loudest thing short of a trap
Ink bomb	Tactical	A blinding cloud, no damage, cuts sight and scent for 8 s
Chum bag	Tactical	Throw a corpse chunk to lure predators to a spot
Flare	Tactical	Light that curious species approach and light-sensitive species flee
Cleaning brush	Equipment	Scrapes parasites off yourself or a teammate
Repair kit	Equipment	Fixes barricade nets faster

Salvage builds (buildables)

Parts are scattered across the map; carry them to a workbench. One build is held at a time.

Build	Parts	Effect
Shell Shield	Turtle shell, strap, brass rim	A back-mounted shield that blocks bites; held in hand for a bash
Turbine	Fan, dynamo, mount	Powers a nearby machine for 90 s before the map's power is on
Net Tripwire	Net, two stakes, bell	Holds the first beast up to size 3 that crosses it
Decoy Buoy	Buoy, lantern, chum tin	Draws beasts and enemies to it for 45 s
Bubble Wall	Compressor, hose, valve	A curtain small beasts won't cross; 60 s

Player stats and progression

- **In-match stats:** kills, headshots, scrip earned, downs, revives, beasts eaten by other beasts because of you, enemies eaten, highest tide.
- **Between matches:** an arcade profile rank from 1 to 50 across all Red Tide maps; ranks unlock charms, skins, and dossier pages. No stat ever changes in-match power; every match starts from the same spawn-in.
- **Dossiers:** every beast, flora, and enemy killed or observed for 30 s adds a dossier page (behavior, diet, weak point, what it fears). The full dossier is the map's real strategy guide.

Player skins

Four divers with fixed silhouettes, each a Depth character type: the Diver (brass helmet), the Whaler (slicker and hood), the Stowaway (a patched suit and a bottle on a cord), and the Mechanic (a riveted hard suit). Skins bought with tokens change suit colors and helmets (verdigris, red tide, bone, pearl, Atlantean) but never the silhouette, so teammates and enemies always read who is who.

Enemy factions

Each map has one faction drawn by the alarm and by Hunts. They are intelligent, they shoot back, and they are also animals in the food web: they bleed, they are hunted, and some of them use the web against the players. Enemies drop their weapons and 200–600 scrip.

Map	Faction	Who they are	Squad (arrives on an alarm roll)	Hunt force	Tactics	In the food web
The Sunken Ship	The Wreckers	Salvage pirates in patched diving suits, here for the yacht's safe	3-4: two Cutters (knives, fast), a Speargunner, a Netman	10 plus a Foreman with a Harpoon Cannon	Flank through the ship's two decks; throw chum at players to call the reef onto them; retreat to the cargo hold	They bleed and panic; sharks and barracuda hunt them; a Wrecker corpse is 20 blood
The Underwater Cave	The Drowned	Ghost sailors from a lost expedition, lanterns on poles	4: three Deckhands (boarding axes), a Lantern Bearer	12 plus the Bosun (a cutlass ghost who can walk through stalagmites)	Advance slowly and steadily; the Lantern Bearer's light draws light-hungry beasts onto the players; they ignore slipstreams	They don't bleed , so predators ignore them, which is why they're dangerous; but the Ghost Worm and cave amphibians attack them as cold things, and their lanterns attract every curious species in the cave
The Coral Reef	Reef Raiders	The Island tribe on a hunting raid: free-divers with spears riding sea turtles, and their Shamans	4: two Spear Divers, a Turtle Rider, a Sling Diver (coconut bombs)	12 plus a Shaman who calls sharks with a drum	Ride turtles through coral corridors; the Shaman's drum raises the aggression of every beast in 40 m for 20 s	They hunt beasts for meat and leave corpses: a raid raises blood everywhere it goes; sharks the Shaman calls don't care whose side anyone is on

Map	Faction	Who they are	Squad (arrives on an alarm roll)	Hunt force	Tactics	In the food web
Atlantis	The Lost Ones	Drowned citizens who breathe water: legionnaires, cultists, and a priest	4: two Legionnaires (gladius and shield), a Cultist (void chant slows players), a Slinger	14 plus an Armored Lost One and a Priest	Phalanx formations in streets; hold plazas; cultists chant to draw the Alien Horror when losing	They bleed black ichor: a different scent that the city's scavengers love and that apex beasts avoid; Lost Ones fight the beasts nesting in their old homes
Approaching the Void	The Remnant	What's left of the research station: mad researchers in pressure suits, clockwork sentinels, and void cultists	4: two Sentinels (rotary needlers, don't bleed), a Researcher (gas grenades, heals sentinels), a Cultist	16 plus the Station Chief in a diving mech	Sentinels hold chokepoints; researchers deploy lure beacons to send beasts at players; cultists try to reach the abyss's edge to summon	Sentinels leak oil (no scent); researchers and cultists bleed normally and are eaten; the Station Chief's mech is size 5 and the abyss's apex beasts attack it on sight

Shared enemy rules

- Enemies see, hear, and smell like beasts: they can be lured by chum, blinded by ink, and distracted by beasts attacking them. Their squad AI prefers a fight against players over beasts but will turn to defend itself.
- Enemies scale with tide: +8% HP per tide, better weapons every 5 tides (Forged weapons at tide 15+).
- An enemy killed by a beast counts for no scrip, but the beast that ate it is now fed and calmer.

- Enemies never respawn indoors near the players; each map has faction entries (a breach in the hull, a cave mouth, the reef's drop-off, the city gate, the station's airlocks) that the players can watch and net.

Shared beast rules

Every beast in every map is built from the same size classes, tiers, and behavior archetypes, so the 200 species are data on top of a small set of systems. Each map's tables give a beast's class, size, tier, diet, predators, and a behavior note; everything else follows from these rules.

Size classes

Size	Scale	HP (tide 1)	Base bounty	Blood on death	Blood per second wounded	Typical speed
1	Finger to hand	20	40	20	2	Fast, erratic
2	Arm	60	80	40	4	Fast
3	Human	250	180	60	6	Diver speed
4	Large (shark, seal)	900	400	80	8	Faster than a diver
5	Huge (great white, whale)	3,000	900	100	10	Fast in a straight line, slow to turn
6	Leviathan (boss)	15,000	1,500	120	12	Slow, unstoppable

HP rises 8% per tide. Bounties rise 10% per tide. Blood does not scale, which is why later tides feel calmer per kill and deadlier per minute.

Tiers

- **Tier 1** grazers, filter feeders, and detritivores: eat flora and corpses; the base of the web and the safest scrip.
- **Tier 2** small predators: eat tier 1; territorial or venomous species here are the early hazards.
- **Tier 3** mid predators: eat tiers 1–2 and each other; the tide 4–9 pressure.
- **Tier 4** apex: eat everything below, including enemies and divers; arrive by blood from tide 4, wander from tide 10.
- **Tier 5** boss: one per map, size 6, a home and a habit; a Predator Hunt can call it.

- **Tags:** scavenger (eats corpses first), parasite (attaches), venom (damage over time on bite or touch), armored (front-facing damage halved), electric (stun), camouflage (invisible until it moves), luminous (draws curious species), amphibious (uses air pockets and dry areas), burrower (hides in sand or silt), cleaner (lowers hosts' aggression).

Behavior archetypes

Archetype	Pattern	Counter
Schooler	Boids; scatter when shot, regroup in 10 s; the school is a blood battery	Melee or one-shot weapons; don't spray
Ambusher	Waits camouflaged in a home spot; strikes anything within 2 m	Flares and Vulture's Eye reveal; walk around, not over
Patroller	Circles a route; investigates blood and sound	Time the route; fight it away from its home
Territorial	Defends a radius; escalates warning, then attack	Leave the radius; use the beast as a wall against enemies
Pack hunter	3–6 coordinate flank and drive; target the wounded and the separated	Backs to a wall; kill the leader (marked)
Swarm	Many bodies, one agent; damage by contact time	Ink, bubble walls, keep moving
Parasite	Attaches to a beast or diver; slows or bleeds until brushed off	Cleaning brush or a cleaner station
Grazer	Eats flora, flees everything	Free scrip, but every corpse is blood
Wanderer	Apex; follows blood from far, leaves when fed	Feed it something else

Damage model

- Weak points: gills, eyes, the soft underside of shelled beasts, the vent of cephalopods, the throat of reptiles. Weak-point hits do 2× and bleed 2×.
- Armored fronts (crabs, turtles, armored fish) halve frontal damage; flank them.

- Nets hold up to size 3; limpet charges and the Cannon Harpoon are the answer for size 4+.
- A beast at 20% HP or less in the Flee state leaves a blood trail; a fleeing beast that reaches its home stops bleeding after 10 s.

What a beast contributes to the web

Killing a beast removes it from its region's population, adds its corpse (a blood source scavengers reduce), and changes who patrols there. Killing all the cleaners in a zone raises every host's aggression; killing all the grazers empties the zone of the predators' reason to come; killing an apex empties a zone of fear for one tide, and then something else moves in.

Level 1: The Sunken Ship

The *Marguerite*, a 60-meter steam yacht on her side on a sand shelf at 30 meters, is the smallest map and the tutorial for the whole game: tight corridors, flora that bites, fish whose blood calls sharks, and an alarm that fills fast when four divers fire in one room.

Layout

Zone	What's there	Opens with
The bridge (start pocket)	Spawn, the Gannet rack, Juggernaut machine visible below	—
Grand salon	An air pocket under the tilted ceiling (the crocodile's lair), Quick Brine, Needler Mk I rack, a workbench	750
Galley and pantry	Flechette 12 rack, the first Locker spot, parasite cleaning station (a cleaner shrimp colony)	1,000
Cabin deck	Six cabins with moray pipes, Speed Brine, Long Speargun rack, Sea-Pattern Carbine rack	1,250
Engine room	Power (the generator), the Pressure Forge, the Goliath's home	1,500

Zone	What's there	Opens with
Stern and swim platform	Bolt Harpoon rack, Double Shot, the second Locker spot, the Wreckers' entry (the hull breach)	1,500
The keel and the sand	Outside: the open shelf under the hull, Kick Brine, the mast with its kelp; sharks' approach lane	2,000

Trap: the cargo crane on the foredeck drops a container across the salon door (1,000).

Hidden quest: the captain's safe (three keys held by the Goliath, the Foreman, and the octopus).

Level mechanics

- **Localized alarm.** The ship is one region with two sub-regions (inside, outside); gunfire inside fills the alarm 2× because the hull carries sound. Splitting two-and-two between inside and outside is the intended lesson.
- **Listing.** Every few tides the wreck settles: a cabin door jams, a container slides, the air pocket shrinks by a third.
- **The air pocket** is the crocodile's lair and the place Wreckers retreat to.
- **Dangerous flora** line the corridors; the price of sprinting is fire coral.

Flora (12)

Flora	Type	Effect
Turf algae	Producer on the hull	Grazer food; parrotfish crunch it loudly, and that crunch draws barracuda
Sea lettuce	Producer	Grazer food in the salon; dies off if grazers are wiped out, then regrows
Fire coral	Hazard	Touch: 15 damage burn and a brief flinch
Stinging hydroids	Hazard	Touch: bleed 3 s; drift in doorways
Barrel sponge	Shelter	Small beasts hide inside; hawksbills eat it
Sea fan	Cover	Blocks sight lines; a moray or stonefish is usually behind one

Flora	Type	Effect
Anemone bed	Host and hazard	Clownfish home; stings divers (10) and holds size-1 beasts
Bladder kelp (on the mast)	Shelter and current	Hides divers from sight; sways with current and shows its direction
Coralline crust	Producer	Snail and sea-hare food
Dead man's fingers	Luminous soft coral	Glowes when touched; draws squid and cuttlefish at night tides
Sargassum raft (surface)	Air and shelter	Shade and shelter outside the hull; jellyfish gather under it
Luminous algae mat (engine room)	Luminous producer	Attracts krill, which attracts sprats, which attracts everything; the Goliath's lamp

Beasts (40)

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Bilge Sprat	Fish	1	1	Krill, algae	Almost everything	Schooler in the corridors; the map's blood battery: 20 sprats is a shark's worth of blood
Silverside	Fish	1	1	Krill	Trevally, barracuda, squid	Schooler outside; flashes when shot, drawing barracuda by sight
Sergeant Major	Fish	1	1	Algae, jellies	Grouper, moray	Guards egg patches on the hull; nips divers (2)
Barnacle Goby	Fish	1	1	Detritus	Sea snake, blenny	Lives in pipes; the sea snake's reason to be in the cabins

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Cleaner Wrasse	Fish	1	Cleaner	Parasites	Nothing willingly	Runs a cleaning station in the galley; hosts there are calm; killing it raises host aggression 20%
Cleaner Shrimp	Crustacean	1	Cleaner	Parasites	Blenny	The other cleaning station; brushes parasites off divers who wait 3 s
Fiddler Crab	Crustacean	1	1	Detritus	Octopus, loggerhead, trevally	Scuttles on the sand outside; waves a claw when a predator is near (a free warning)
Hermit Crab	Crustacean	1	Scavenger	Corpses	Octopus	Reduces corpse blood; killing them slows blood cleanup
Whelk	Mollusc	1	1	Coralline crust	Crabs, loggerhead	Slow; free scrip
Sea Hare	Mollusc	2	1	Sea lettuce	Nothing (toxic)	Inks when shot: a free ink cloud
Feather Duster Worm	Worm	1	1	Plankton	Blenny	Retracts when a diver passes; a motion sensor for ambushers
Rudder Krill	Crustacean	1	1 (swarm)	Plankton, algae mat	Sprats, whale-sized nothing here	A swarm around the algae mat; shooting it is pointless and loud
Parrotfish	Fish	2	1	Turf algae	Grouper, bull shark	Crunches loudly (sound layer); sleeps in a mucus cocoon in cabins at night tides
Rabbitfish	Fish	2	1	Algae	Barracuda	Venomous dorsal spines: touching a live one hurts

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Damselfish	Fish	1	2	Small invertebrates	Trevally	Territorial; attacks divers in its patch; a swarm of bites at 1 each
Blenny	Fish	1	2	Worms, gobies	Moray, sea snake	Pipe-dweller
Arrow Crab	Crustacean	1	2	Worms	Wrasse, octopus	On sea fans; harmless
Mantis Shrimp	Crustacean	2	2	Crabs, snails	Octopus, grouper	Punches: 30 damage and breaks a Shell Shield in two hits; burrows in the sand
Lionfish	Fish	2	2	Sprats, gobies	Grouper (rarely)	Venom on touch (poison 5 s); drifts through doorways
Squirrelfish	Fish	2	2	Crabs	Moray	Night tides only; large eyes, loud clicks
Reef Squid	Cephalopod	2	2	Sprats, silversides	Trevally, barracuda, tuna	School; inks when threatened; drawn to dead man's fingers
Cuttlefish	Cephalopod	2	2	Crabs, shrimp	Grouper	Camouflaged in cabins; hypnotic flashing slows a diver's aim briefly
Banded Sea Snake	Reptile	2	2	Gobies, blennies, eels	Tiger shark	Hunts the pipes; must surface at the air pocket every 90 s (predictable); venom: 20 poison over 10 s
Stonefish	Fish	2	2	Sprats	Nothing	Ambusher on the sand and in the galley; step on it: 40 damage and poison
Jelly Bloom	Cnidarian	2	1 (swarm)	Plankton	Turtles	Drifts under the sargassum and into the stern; contact stings 5 per second

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Remora	Fish	1	Parasite	Scraps	Barracuda	Attaches to sharks, turtles, and divers; on a diver, slows swim 15% until brushed off
Lamprey	Fish	2	Parasite	Blood	Barracuda	Attaches and bleeds 2 per second; its host becomes a blood source that draws predators
Green Moray	Fish	3	3	Octopus, squirrelfish, blennies	Grouper, bull shark	Ambusher in cabin pipes; strikes 2 m from its hole, then retreats; territorial
Great Barracuda	Fish	3	3	Silversides, squid, sprats	Bull shark, tiger shark	Patroller in the salon and outside; drawn by flashes and the parrotfish crunch; hits and runs
Grouper	Fish	3	3	Parrotfish, lionfish, morays, cuttlefish	Bull shark	Ambusher in the engine room's shadows; swallows size 1 whole
Bluefin Trevally	Fish	3	3 (pack)	Silversides, squid, damselfish	Sharks	Pack of 5 outside the hull; drives schools against the hull where divers stand
Common Octopus	Cephalopod	3	3	Crabs, mantis shrimp, snails	Moray, grouper	Steals a diver's held item (a part or a charm) and hides in a cabin; kill it to get it back
Nurse Shark	Fish	3	Scavenger	Corpses, crabs	Tiger shark	Night tides; docile; eats corpses fast (the divers' best friend)

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Hawksbill Turtle	Reptile	3	2	Sponges, jellies	Tiger shark	Armored; eats the jelly bloom; harmless unless shot
Loggerhead Turtle	Reptile	4	3	Crabs, whelks, mantis shrimp	Tiger shark, crocodile	Armored; crushes a diver's shield in a bite if provoked
Sand Tiger Shark	Fish	4	4	Trevally, barracuda, grouper	Tiger shark	Hangs motionless near the keel; investigates blood over 60
Bull Shark	Fish	4	4	Everything size 3 and below, Wreckers	Tiger shark, crocodile	Enters the hull through the breach; aggressive; threshold 80 blood
Saltwater Crocodile	Reptile	5	4	Turtles, bull sharks, Wreckers, divers	Nothing but the Goliath	Amphibious: lives in the salon's air pocket; a death roll grabs a diver (teammates shoot it off); leaves for the sand when the pocket shrinks
Tiger Shark	Fish	5	4	Everything, including sea snakes and turtles	Nothing	Wanderer; threshold 150 blood; eats enemies as readily as divers; leaves when fed
The Goliath	Fish	6	5	Anything that enters the engine room	—	A goliath grouper the size of a boiler: inhales a diver from 4 m (teammates have 5 s to shoot it out); guards the Forge; wanders from tide 12

The web in motion

- **The sprat chain:** algae mat → krill → sprats → trevally and barracuda → bull shark → tiger shark. Shooting sprats for cheap scrip is the map's first trap.
- **The cleaner rule:** the galley and the outside cleaning stations calm the hosts that visit (groupers, barracuda, turtles). Wiping them out makes the mid tiers hostile everywhere.
- **The crocodile problem:** the only air inside is its lair. Feed it a turtle (shoot one near the pocket) and it eats for 60 s.
- **Wreckers feed the reef:** a Wrecker squad dying in the salon is 80 blood in a closed room; the bull shark arrives 30 s later. Let the reef kill them outside instead.
- **The Goliath:** everything that flees runs to the engine room's shadows and gets eaten there. The Forge is safe only right after it has fed.

Definition of Done

- All 40 beasts and 12 flora exist as data and models; `--eco-sim ship 20` shows the sprat chain calling a bull shark within 90 s of 15 sprat kills in the salon.
- A four-player LAN match reaches tide 10 with the crocodile, a Wrecker Hunt, and the Goliath's inhale all seen.

Level 2: The Underwater Cave

A flooded cave system with dry air chambers, a deep sump, and six slipstreams that carry divers and beasts between its halls in seconds. It is dark, loud (every shot echoes), and half its cast is insects and amphibians that live where the water meets the air.

Layout

Zone	What's there	Opens with
The Mouth (start pocket)	Spawn under a light shaft, the Gannet rack, Juggernaut	—
The Flooded Gallery	Long hall of columns; Needler Mk I, Quick Brine, workbench; slipstream A to the Chimney	750

Zone	What's there	Opens with
The Chimney	A vertical shaft; ledges with Long Speargun; stalactites overhead (the map's trap: shoot the anchor to drop them, 1,000)	1,000
The Dry Chambers	Air, three linked caverns with pools; insects and amphibians; Speed Brine, Carbine rack; the Drowned's entry	1,250
The Sump	The deep, silty bottom; Bolt Harpoon, Double Shot; the Sleeper Shark's home; bacterial mats	1,500
The Dynamo Sump	Power: a drowned expedition's generator; the Forge; slipstreams C and D converge here	1,750
The Cathedral	A vast chamber with the crystal coral and the Lobster's home; Kick Brine, second Locker spot	2,000

Slipstreams A–F are one-way currents (8 m/s) between zones; a diver can't fire inside one and arrives with 2 s of drift. Beasts use them too: what you see enter a slipstream behind you arrives where you do. The Drowned do not use them.

Level mechanics

- **Rockfall.** Gunfire in the Chimney and the Cathedral has a 5% chance per loud shot (noise 6+) to drop a stalactite somewhere nearby: 200 damage to whatever is under it, beast, enemy, or diver. The Resonator drops them on purpose.
- **Air chambers.** The Dry Chambers hold the map's insects and amphibians; divers walk there, slowly, and the toad's tongue reaches from the pool.
- **Echo.** The cave counts as one region for the alarm; the Drowned come from the Dry Chambers.
- **Bloodvine.** A red alga that smells like blood to the web: shooting a patch makes predators investigate it. The cave's built-in decoy, and its built-in trap when it grows near the Forge.

Flora (12)

Flora	Type	Effect
Glow mould	Luminous producer	The base of the wet web; lights corridors; cavefish graze it
Chemosynthetic mat (sump vents)	Producer	The base of the sump web; krill and isopods swarm it
Pale cave moss (dry chambers)	Producer	Insect and newt food
Snottite curtain	Hazard	Acid drips: 5 damage per second under it
Crystal coral (Cathedral)	Hazard and tool	Sharp (touch bleeds); shooting a cluster rings it, calling the Lobster
Root tangle	Shelter	Hanging roots from above; divers hide in it; crayfish nest in it
Drip fungus	Hazard	A spore cloud when disturbed: slows 30% for 5 s
Ink cap	Tool	Bursts into an ink cloud when shot
Bloodvine	Tool and trap	Smells like 40 blood when cut
Cave sponge	Shelter	Glass shrimp and blennies inside
White anemone cluster	Hazard	Holds size 1 and stings divers (8)
Stone lettuce	Producer	Sturgeon and snapping turtle food

Beasts (40)

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Blind Cavefish	Fish	1	1	Glow mould	Eels, catfish, anglerfish	Schooler; no eyes, so it ignores light and follows sound

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Glass Shrimp	Crustacean	1	1	Detritus	Everything small	Transparent; lives in sponges; the cleaner of this map
Cave Krill	Crustacean	1	1 (swarm)	Bacterial mat	Cavefish, sturgeon, gulper	Sump swarm
Cave Isopod	Crustacean	1	Scavenger	Corpses	Crayfish	Silt scavengers; clean corpses slowly
Amphipod	Crustacean	1	1	Detritus	Newts, beetles	Pools and puddles in the dry chambers
Troglobite Crab	Crustacean	2	2	Amphipods, glass shrimp	Caiman, toad	Ghost-white; clicks in the dark
Olm	Amphibian	2	2	Amphipods, isopods	Giant salamander	Blind; senses electricity; it can feel a diver's rebreather through a wall
Blind Newt	Amphibian	2	2	Insects, amphipods	Toad, krait	Pools in the dry chambers
Poison Dart Newt	Amphibian	1	2	Crickets	Nothing (toxic)	Bright even in the dark; touch: 10 poison
Giant Cave Toad	Amphibian	3	3	Newts, crickets, beetles, divers' hands	Caiman	Sits at the pool's edge; a tongue grab from 5 m pulls a diver into the pool
Cave Crayfish	Crustacean	2	2	Isopods, worms	Catfish, salamander	Nests in root tangles; pinches divers who hide there
Fishing Spider	Insect (arachnid)	2	2	Cavefish, newts	Swiftlets	Runs on the water's surface in the chambers; drops onto divers from roots

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Cave Cricket	Insect	1	1 (swarm)	Cave moss	Newts, spiders, swiftlets	Chirp swarm; goes silent when the Drowned pass (a warning)
Diving Beetle	Insect	1	2	Amphipods, newt larvae	Newts	Carries a bubble; harmless, free scrip in pools
Giant Water Bug	Insect	2	2	Newts, small fish	Toad	Pierces a diver's suit: 15 bleed
Dragonfly Nymph	Insect	2	2	Cavefish, amphipods	Toad, newts	Jet-propelled; a lunging jaw
Glowworm Colony	Insect	1	1	Cave crickets, midges	Nothing	Ceiling threads that glow; curious species gather beneath; lights the Chimney
Firefly Squid	Cephalopod	2	2	Cavefish, krill	Eels, anglerfish	Luminous school; flashes when threatened, lighting divers up for ambushers
Lanternfish	Fish	2	1	Krill	Gulper, cave eel	Sump school; its lights draw the gulper
Cave Anglerfish	Fish	3	3	Cavefish, squid	Sleeper shark	A lure that looks like a drop bottle from a distance
Cave Eel	Fish	3	3	Cavefish, squid, crayfish	Caiman, sleeper shark	Ambusher in wall cracks; uses slipstreams to relocate
Electric Ray	Fish	3	3	Crayfish, cavefish	Sleeper shark	Buried in silt; a shock stuns divers for 2 s and stops a Turbine
Blind Catfish	Fish	3	3	Crayfish, cavefish, isopods	Caiman	Whiskers find blood fast: the first mid predator to any corpse

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Whip Ray	Fish	3	3	Crabs, shrimp	Sleeper shark	Glides the Gallery; a tail barb (bleed) if stepped on
Hagfish	Fish	2	Scavenger	Corpses	Nothing (slime)	Slimes divers who touch a corpse it's in (slows 40% until brushed)
Vampire Squid	Cephalopod	2	Scavenger	Marine snow, corpse bits	Gulper	Sump drifter; harmless; its cloak flashes when shot
Bristle Worm	Worm	2	Scavenger	Corpses, mats	Catfish	Venomous bristles: touch burns 10
Sea Spider	Crustacean-like (pycnogonid)	2	2	Anemones, sponges	Crayfish	Slow; hollow; ignores divers
Cave Swiftlet	Bird	1	2	Crickets, spiders	Nothing here	Flies in the dry chambers; a flock startled by gunfire is a loud alarm
Snapping Turtle	Reptile	3	3	Newts, fish, stone lettuce	Caiman	Armored; sits still; a bite takes 60 and holds
Cave Krait	Reptile	2	2	Newts, cavefish	Snapping turtle	Swims between pools; venom 25 over 10 s; surfaces to breathe in the chambers
Epaulette Shark	Fish	3	3	Crabs, worms	Caiman	Walks on its fins across dry rock between pools; ambushes from ledges

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Current Runner	Fish	3	3	Cavefish, squid	Sleeper shark	A sailfish-like hunter that lives in the slipstreams; hits divers riding them
Cave Sturgeon	Fish	4	1	Krill, stone lettuce, mats	Sleeper shark	Armored grazer; harmless; a huge blood source if killed (the sump's trap)
Gulper Eel	Fish	4	3	Lanternfish, squid, vampire squid	Sleeper shark	Sump; swallows size 2 whole; a bite on a diver takes 80 and pins them 3 s
Giant Salamander	Amphibian	4	4	Olms, crayfish, catfish, divers	Caiman	Ambusher under the pool ledges in the dry chambers; a bite holds and drags
Cave Caiman	Reptile	4	4	Toads, turtles, salamanders, the Drowned's flesh (none), divers	Sleeper shark	Amphibious patroller between the chambers and the Gallery; threshold 60 blood
Sleeper Shark	Fish	5	4	Everything in the sump, caiman, enemies	Nothing	Slow, silent, cold; threshold 120 blood; nearly invisible in silt; a bite takes 150
Ghost Worm	Worm	4	4	The Drowned, eels	Nothing	From Depth's Cave: lives in the walls; attacks the Drowned on sight and any diver who shoots it; the cave's answer to the faction that doesn't bleed

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Silt Lamprey	Fish	1	Parasite	Blood	Catfish	Attaches in the sump; bleeds 1 per second
The Lobster	Crustacean	6	5	Anything in the Cathedral	—	From Depth's Cave: a boiler-sized lobster; claws crush a shield outright and hold a diver; its shell halves damage from the front; the crystal coral calls it

The web in motion

- **Two webs, one cave.** The wet web (mould → cavefish → eels and catfish → caiman → sleeper shark) and the dry web (moss → crickets → newts and spiders → toad → caiman and salamander) meet at the pools, and the caiman walks both.
- **The Drowned don't bleed**, so only the Ghost Worm and the caiman (which attacks cold things by habit) fight them. A team that learns to lead the Drowned past the Ghost Worm's wall crack wins Hunts cheaply.
- **Lanterns and lights:** the Lantern Bearer, the glowworms, firefly squid, and drop bottles all draw curious species. The anglerfish's lure looks like a drop.
- **The sturgeon:** 80 blood for one harmless fish. New players shoot it; the sleeper shark thanks them.
- **Slipstreams move blood:** a wounded fish fleeing into a slipstream carries its trail to the next hall.

Definition of Done

- Slipstreams carry divers and beasts; `--eco-sim cave 20` shows a wounded eel's trail crossing halls and a caiman following it.
- Rockfall, the toad's tongue grab, the sturgeon trap, and a Drowned Hunt led into the Ghost Worm all occur in a four-player test.

Level 3: The Coral Reef

The bright map: sunlit coral forests whose corridors hide eels and grabbing coral, a lagoon edged with mangroves where frogs and toads live, a sand flat with its own cast, and a drop-off where dolphins, seals, and whales pass. Mammals and amphibians join the fish, reptiles, and crustaceans, and the reef's first lesson is that some of them are on your side until you shoot one.

Layout

Zone	What's there	Opens with
The Lagoon (start pocket)	Shallow, bright; mangrove roots on one edge; Gannet rack, Juggernaut	—
Staghorn Forest	Dense coral corridors; eel holes; Needler Mk I, Quick Brine, workbench	750
The Bommie	A giant coral head with a cleaning station on top; power (a tide mill in the surge channel); Speed Brine	1,000
The Sand Flats	The open expanse: seagrass meadows, garden eels, rays, the bobbit worm; Long Speargun, Bolt Harpoon, Double Shot; slow sightlines, no cover	1,250
Brain Coral Maze	Winding corridors of brain and table coral; Reacher coral gates; Flechette, Carbine racks; the Raiders' entry from the drop-off	1,500
The Blue Hole	A deep sinkhole in the flats; the Forge on a ledge; the hammerhead's and the whales' route; Kick Brine	2,000
The Drop-off Wall	Open blue; mantas and the orca pod; second Locker spot; Mule Brine	2,000

Traps: a Reacher coral gate (1,000) closes a corridor and eats whatever it holds; the surge channel (1,000) flushes the Bommie's top. Hidden quest: return the Shaman's drum to the sea turtle nesting beach.

Level mechanics

- **Coral corridors** limit fire lanes to 8–12 m; loud guns are wasted, spearguns and knives shine.
- **Eel holes** line the corridors: passing within 1 m risks a moray or ribbon eel strike; Vulture's Eye shows occupied holes.
- **Reacher coral** grabs a diver who brushes it and holds until a teammate cuts them loose (or the coral is shot to 0); it also grabs beasts and enemies, which feed it.
- **The sand flats** have no cover and full sightlines: the loud-gun zone, and the place where a downed diver's blood spreads fastest.
- **Dolphins.** The pod follows the divers and drives sharks off while nobody has shot a dolphin. One shot turns the pod hostile for the match.
- **Crown-of-thorns** eat coral: leaving them alive erodes corridors (opening lanes for sharks); killing them is 40 blood each.

Flora (12)

Flora	Type	Effect
Staghorn thicket	Structure and producer	The corridors themselves; grazed by butterflyfish and crown-of-thorns; breaks under a shark ram
Brain coral	Cover	Solid; blocks shots and sight
Table coral	Shelter	Roofs that hide divers from above (mantas, sharks)
Fire coral	Hazard	Touch: 15 burn
Reacher coral	Hazard and trap	Grabs and holds; feeds on what it holds
Seagrass meadow	Producer (flats)	Dugong and turtle food; cover for stingrays; hides the bobbit worm's hole
Mangrove roots	Shelter (lagoon)	Air pocket above; the frog and toad nursery; juvenile fish hide here, which draws barracuda
Sea whip fans	Cover	Sway with current; seahorses and pipefish cling to them
Soft coral garden	Producer	Angelfish food; luminous at night tides

Flora	Type	Effect
Halimeda beds	Producer	Sea hare and urchin food; brittle: shatters loudly when walked on
Turf algae	Producer	Surgeonfish and iguana food
Sea grape	Edible	A diver can eat a cluster: +10 HP, once per cluster; grazers compete for it

Beasts (41)

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Anthias	Fish	1	1	Plankton	Groupers, triggerfish, jacks	A shimmering school over the Bommie; the reef's blood battery
Clownfish	Fish	1	1	Plankton, algae	Nothing while in its anemone	Bites divers near its anemone (1 damage, endlessly)
Butterflyfish	Fish	1	1	Coral polyps	Groupers	Grazes staghorn; harmless
Cleaner Wrasse	Fish	1	Cleaner	Parasites	Nothing willingly	The Bommie's cleaning station calms everything that visits, including sharks
Sea Urchin	Echinoderm	1	1	Halimeda, turf	Triggerfish, wrasse	Spines: 8 damage on touch
Spiny Lobster	Crustacean	2	2	Urchins, snails	Groupers, octopus, seals	In crevices; free scrip at night tides
Angelfish	Fish	2	1	Sponges, soft coral	Reef sharks	Pairs; harmless
Triggerfish	Fish	2	2	Urchins, crabs	Reef sharks	Territorial over nests; bites (10) and chases
Surgeonfish	Fish	2	1	Turf algae	Reef sharks	Scalpel tail: 12 bleed if grabbed

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Garden Eel	Fish	1	1	Plankton	Stingrays	Hundreds in the flats' sand; retract when divers approach, which shows a diver's position to everything watching
Goatfish	Fish	2	2	Worms, crabs	Jacks, sharks	Stirs sand in the flats; a visible dust cloud
Flounder	Fish	2	2	Shrimp, small fish	Stingrays, sharks	Camouflaged on sand; startles when stepped on
Razorfish	Fish	1	1	Plankton	Jacks	A vertical school that hides among sea whips
Sand Dollar	Echinoderm	1	1	Detritus	Nothing	Free scrip; a carpet in the flats
Cone Snail	Mollusc	1	2	Small fish, worms	Nothing	Harpoon sting on touch: 30 poison over 15 s
Blue-ringed Octopus	Cephalopod	1	2	Crabs, shrimp	Nothing	Rings flash when threatened; a bite is 60 poison over 20 s and paralysis 3 s; the reef's deadliest small thing
Box Jelly	Cnidarian	2	2	Shrimp, small fish	Green turtle	Nearly invisible; contact 25 and paralysis 1 s
Crown-of-thorns Starfish	Echinoderm	2	1	Coral	Triton snail (none here), nothing	Erodes corridors; spines: 10 poison
Ribbon Eel	Fish	2	2	Small fish	Grouper	Eel-hole ambusher; a quick bite (20)
Reef Cuttlefish	Cephalopod	2	2	Crabs, shrimp	Grouper, dolphins	Camouflage; dazzles a diver's aim
Crab-eating Frog	Amphibian	2	2	Crabs, insects	Sea kraits, toad	Mangrove roots; leaps between roots and the surface

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Reef Axolotl	Amphibian	2	2	Shrimp, worms	Toad, groupers	Lagoon pools; regenerates: heals 5 per second when hidden, so it's a tough small kill
Giant Reef Toad	Amphibian	3	3	Frogs, axolotls, crabs	Monk seal, sharks	Mangrove edge; tongue grab from 5 m; croaks are a sound-layer alarm
Marine Iguana	Reptile	2	1	Turf algae	Sharks	Grazes on the Bommie's shallows; harmless; a herd is 200 blood if wiped
Olive Sea Snake	Reptile	2	2	Eels, gobies	Sharks	Curious; approaches divers; venom 25 over 10 s
Green Turtle	Reptile	3	1	Seagrass, jellies	Hammerhead, orca	Armored; eats box jellies (a reason to keep it alive)
Moray Eel	Fish	3	3	Octopus, cuttlefish, lobsters	Groupers, sharks	In the corridor holes; strikes 2 m; territorial
Coral Grouper	Fish	3	3	Anthias, butterflyfish, eels	Sharks	Hunts with the moray (a real partnership): the moray flushes, the grouper catches
Stingray	Fish	3	3	Garden eels, flounder, goatfish	Hammerhead	Buried in seagrass; barb 40 bleed if stepped on
Bobbit Worm	Worm	3	3	Anything that passes its hole	Nothing	Sand flats ambusher: a 3 m lunge from a hidden hole, 80 damage and a drag; the flats' reason to follow the goatfish clouds instead of the shortcuts

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Titan Triggerfish	Fish	3	3	Urchins, crabs, coral	Sharks	Defends a cone-shaped territory above its nest; charges divers at 3× speed
Napoleon Wrasse	Fish	4	2	Urchins, crustaceans, cone snails	Orca	Enormous and gentle; eats cone snails and blue-rings; 80 blood if killed
Dugong	Mammal	4	1	Seagrass	Orca, hammerhead	Grazes the flats; harmless; the flats' sturgeon trap
Blacktip Reef Shark	Fish	4	4 (pack)	Fish, cuttlefish, iguanas	Hammerhead, orca	Packs of 4 in the corridors; threshold 60; skittish alone
Grey Reef Shark	Fish	4	4	Fish, octopus, seals' pups	Orca	Patrols the wall; threshold 80; a threat display (hunched swim) before attacking
Bottlenose Dolphin	Mammal	4	3	Fish, squid	Orca	A pod of 5 that shadows the divers and rams sharks off them; hostile forever after one dolphin is shot
Monk Seal	Mammal	4	3	Fish, lobsters, octopus	Hammerhead, orca	Curious; steals a held item and plays keep-away; harmless unless shot
Manta Ray	Fish	5	1	Plankton	Orca	Glides the wall and the blue hole; harmless; 100 blood if killed
Great Hammerhead	Fish	5	4	Stingrays, sharks, turtles, seals, enemies	Orca	Wanderer; threshold 150; electroreception finds buried divers and buried rays

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Humpback Whale	Mammal	5	1	Krill (passes through)	Nothing	Passes the wall every few tides; harmless; a diver who shoots it produces 100 blood and a Predator Hunt on the spot
The Matriarch	Mammal	6	5	Sharks, seals, dolphins, dugongs, enemies, divers	—	An orca matriarch with a pod of 3: they hunt as one, herd divers away from cover, and ram the Bommie; the Shaman's drum can call her, which is the Raiders' mistake to exploit

The web in motion

- **The corridor pair:** moray flushes, grouper catches; kill one and the other becomes reckless.
- **Dolphins versus sharks:** the pod's protection is worth more than any tonic; the Raiders know it and shoot dolphins on purpose to turn them against the divers.
- **The flats:** garden eels retract (revealing you), goatfish clouds mark safe sand, the bobbit worm punishes the straight line, and the dugong's blood calls the hammerhead.
- **Keep the eaters of dangerous things alive:** green turtles eat box jellies; Napoleon wrasse eat cone snails and blue-rings.
- **The drum:** the Shaman's drum raises aggression and can call the Matriarch. Kill the Shaman first, or take the drum and use it on the Raiders.

Definition of Done

- Reacher coral grabs and feeds; the dolphin alliance flips on a shot; `--eco-sim reef 20` shows crown-of-thorns eroding a corridor within 10 minutes if unchecked.
- A four-player test sees a Raider Hunt, the flats at night, and the Matriarch's pod.

Level 4: Atlantis, the Forgotten City

A walled city on a hill, drowned and overgrown: farms and irrigation on the flat ground inside the wall, workshops and markets in the lower town, bathhouses and barracks on the slopes, and the government hall and the grand chapel on the summit. Every building type is now a habitat, and the city's own water systems, the aqueducts and cisterns, move blood and beasts through it.

Layout (a real city plan)

District	Buildings	Ecology	What's there	Opens with
The Harbor Gate (start pocket)	The gatehouse in the great wall, a customs house	Sardines in the gate arch; the Lost Ones' entry	Spawn, Gannet rack, Juggernaut	—
The Farms	Terraced fields, irrigation canals, granaries, a mill	Seagrass meadows now; the crocodile's canals; grazers	Needler Mk I, Quick Brine, workbench, canal traps	750
The Lower Town	Houses, food stores, a market square, the toolsmiths' street	Octopi in amphorae, eels in drains, cardinalfish in cellars	Carbine and Flechette racks, Speed Brine, Locker spot 1, workshop parts	1,000
The Bathhouses	Hot and cold baths, a gymnasium, latrines	Sponges and groupers in the baths; the water monitor in the dry hypocaust rooms	Double Shot, Long Speargun; an air pocket in the caldarium dome	1,250
The Barracks and Armory	Drill yard, armory, stables, a small fort on the wall	The Lost Ones' garrison; amberjacks drill the yard	Trawlerman's Rifle, Bolt Harpoon, Mule Brine; the Lost Ones' Hunt staging	1,500

District	Buildings	Ecology	What's there	Opens with
The Library and Academy	Reading halls, scroll stores, an observatory	Dark stacks: the giant squid; glow-moss on the shelves	Kick Brine, the second workbench, Locker spot 2	1,750
The Forum and Government Hall (summit)	Council chamber, treasury, courts	The forum's sea fans; tuna and swordfish cross the open plaza	Power: the treasury's crystal engine; Deadeye Brine	2,000
The Grand Chapel (summit)	Nave, crypts, the god-pool	The Cistern Wyrms' pool; temple kraits in the crypts	The Pressure Forge; Widow's Ink	2,500

The city has 60+ buildings on a hill 120 m tall and 500 m across, connected by streets, stairs, and the aqueduct. Traps: a canal sluice (floods a farm terrace), the bathhouse hypocaust (boils a room), the chapel's god-pool (drains and refills). Hidden quest: rekindle the lighthouse on the wall to see the whole city on the sonar.

Level mechanics

- **The aqueduct carries blood downhill.** Blood made on the summit flows down the aqueduct into the baths and the lower town within a minute. Fighting uphill sends predators to the players who stayed below.
- **Cisterns and drains.** A network under every district: conger eels and the Wyrms come out of any drain grate; grates can be netted (a build) or watched.
- **Buildings are homes.** Every building type hosts its species by data: amphorae mean octopus, baths mean grouper, cellars mean cardinalfish, granaries mean bream. The dossier teaches which door has what behind it.
- **The Lost Ones live here.** Squads patrol streets, hold plazas, and know the drains. Their ichor scent draws the city's scavengers and repels its apex beasts, so a street full of dead Lost Ones is briefly safe from sharks and full of crabs.
- **The open plaza** on the summit is the only long sightline: the tuna and swordfish lane, and the Priest's chosen ground.

Flora (12)

Flora	Type	Effect
Posidonia meadows (farms)	Producer	Grazer base for the whole lower city
Sea lettuce terraces	Producer	Farm grazers; regrows fast
Column coral	Structure	Grows on every pillar; sharp corners bleed
Doorway kelp curtains	Cover	Hangs in every doorway; hides what's inside, both ways
Bathhouse sponges	Shelter	Cardinalfish and shrimp inside; groupers wait beside them
Forum sea fans	Cover	Break sightlines across the plaza
Temple glow-moss	Luminous producer	Lights the chapel and the library; draws squid
Amphora anemones	Hazard	Stings (10); an octopus is usually behind one
Barnacle marble	Hazard	Cuts (bleed 3 s) on every wall a diver brushes at speed
Fire coral on the wall	Hazard	Burns (15) along the rampart walk
Courtyard sargassum	Air	Air pockets in the roofless courtyards; man o' war beneath
Stinging hydroid drains	Hazard	Every drain grate stings (bleed) when touched

Beasts (40)

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Sardine	Fish	1	1	Plankton	Tuna, amberjacks, dentex	Street schools; the city's blood battery
Cardinalfish	Fish	1	1	Plankton	Groupers, sea bass	Cellars and sponges; nocturnal

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Cleaner Goby	Fish	1	Cleaner	Parasites	Nothing willingly	Cleaning stations at the bath drains
Snapping Shrimp	Crustacean	1	2 (swarm)	Small worms	Bream, octopus	Colonies in the walls; their pops are constant sound-layer noise that hides gunfire partially (a quiet-gun bonus in the lower town)
Porcelain Crab	Crustacean	1	1	Plankton	Octopus, monitor	In anemones; free scrip
Nudibranch	Mollusc	1	1	Sponges, hydroids	Nothing (toxic)	Bright; poison on touch
Sea Cucumber	Echinoderm	1	Scavenger	Detritus	Nothing	Cleans corpses slowly; spits sticky threads if shot
Feather Star	Echinoderm	1	1	Plankton	Nothing	On the columns; harmless
Long-spined Urchin	Echinoderm	1	1	Algae	Triggerfish, monitor	Carpets the lower town's stairs
Bream	Fish	2	1	Algae, snapping shrimp	Dentex, sea bass	Granary schools
Salema	Fish	2	1	Posidonia	Amberjacks	Farm herds; harmless; eating a hallucinogenic patch makes them erratic (the map's joke and a sign of which meadow is off-limits)
Scorpionfish	Fish	2	2	Cardinalfish, shrimp	Groupers	Ambusher on marble; venom (poison 8 s)
Cuttlefish	Cephalopod	2	2	Crabs, shrimp	Sea bass	Market square; dazzles
Slipper Lobster	Crustacean	2	2	Snails, worms	Octopus, monitor	Under the market stalls

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Mantis Shrimp	Crustacean	2	2	Crabs, snails	Octopus	Breaks shields; nests in the toolsmiths' anvils
Aqueduct Newt	Amphibian	1	2	Insects, shrimp	Cistern frogs, kraits	Aqueduct channels; harmless
Cistern Frog	Amphibian	2	2	Newts, shrimp	Monitor, conger	Cisterns; croaks echo through drains (a sound-layer alarm carrier)
Temple Krait	Reptile	2	2	Eels, gobies	Sea bass	Crypt sea snake; venom 30 over 10 s; surfaces at the caldarium air pocket
Water Monitor	Reptile	3	3	Crabs, frogs, urchins, eggs	Crocodile	Amphibious; prowls the bathhouse dry rooms and the drill yard; bites hold
Cormorant	Bird	2	2	Sardines, bream	Nothing	Dives from the courtyard air pockets; a startled flock is an alarm
Moon Jelly Bloom	Cnidarian	2	1 (swarm)	Plankton	Leatherback	Drifts the forum; mild stings
Man o' War	Cnidarian	2	2	Small fish	Leatherback, loggerhead	Under courtyard sargassum; tentacles 10 m long; 30 damage and paralysis 1 s
Octopus	Cephalopod	3	3	Crabs, lobsters, shrimp	Conger, dentex, monk seal	One per amphora street; steals items; the toolsmiths' parts are usually in its den
Conger Eel	Fish	3	3	Octopus, frogs, cardinalfish	Crocodile, great white	The drains; strikes from grates; retreats into the cistern

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Spider Crab	Crustacean	3	2	Sea cucumbers, algae	Octopus (rarely)	Armored; decorates itself with kelp and looks like a bush until it moves
Sea Bass	Fish	3	3	Bream, cuttlefish, kraits	Sharks	Bathhouse patroller
Dentex	Fish	3	3	Sardines, bream, octopus	Sharks, tuna	A fast street hunter in pairs
Amberjack	Fish	3	3 (pack)	Sardines, salema	Sharks	A pack of 6 that drills the barracks yard in formation
Barracuda	Fish	3	3	Sardines, cormorants	Sharks	Waits in the gate arch; drawn to flashes
Lamprey	Fish	2	Parasite	Blood	Barracuda	Cistern parasite; bleeds hosts
Bluefin Tuna	Fish	4	3	Sardines, amberjacks	Great white	Crosses the summit plaza at full speed; a ram is 60 damage
Swordfish	Fish	4	3	Sardines, cuttlefish	Great white	Plaza lane; a charge impales for 100 and pins
Loggerhead	Reptile	4	3	Man o' war, crabs	Crocodile, great white	Armored; eats the man o' war under the sargassum
Monk Seal	Mammal	4	3	Octopus, lobsters, bream	Great white	Curious thief; harmless unless shot
Sandbar Shark	Fish	4	4	Sea bass, dentex, octopus, Lost Ones (reluctantly)	Great white	Patrols the lower town streets; threshold 70
Leatherback	Reptile	5	1	Jellies, man o' war	Nothing	Gentle; 100 blood; eats every jelly in the forum if left alone

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Nile Crocodile of the Canals	Reptile	5	4	Monitors, loggerheads, frogs, Lost Ones, divers	Nothing but the Wyrms	Farms' canals; ambushes from the sluice gates; hauls out on the terraces
Great White	Fish	5	4	Tuna, seals, sharks, enemies, divers	Nothing but the Wyrms	Wanderer over the wall; threshold 150; avoids ichor-scented streets
Giant Squid	Cephalopod	5	4	Tuna, seals, squid, anyone in the stacks	Nothing	The library's dark: reaches through doorways; the glow-moss shows its eye
The Cistern Wyrms	Fish	6	5	Anything near a drain	—	A conger the length of a street living in the cistern network: surfaces from any grate, drags a diver into the drains (teammates have 6 s), guards the chapel's god-pool; from tide 12 it hunts the streets

The web in motion

- **Downhill:** the aqueduct means the summit's fights are the lower town's problem; a team that sets one diver below to hold the drains will lose them to the Wyrms.
- **Ichor:** dead Lost Ones repel sharks and the Wyrms for 60 s and draw crabs, cucumbers, and the monitor; a street of their dead is a safe corridor for a minute.
- **The library:** the squid can't leave the stacks; the Locker there is a trade of pulls for reach.
- **The plaza lane:** tuna and swordfish cross at speed; standing in the lane is a ram; standing beside it and shooting a tuna brings the great white over the wall.
- **The farms:** harmless grazers everywhere, and the crocodile in every canal; the sluice trap floods a terrace and drowns a Lost One phalanx.

Definition of Done

- The city generator places 60+ buildings by district with correct habitats; the aqueduct moves blood downhill in `--eco-sim atlantis 20`.
- The Wurm surfaces from grates; ichor repels and attracts as specified; a four-player test reaches the chapel and survives a Lost Ones Hunt on the summit.

Level 5: Approaching the Void

The largest map: a black sand plain on the rim of an abyss, and beneath it a cave system, built as vertical chimneys and galleries rather than the second map's halls, that descends past a drowned research station to a vault that looks straight down into the void. The rim is empty and enormous; the guns, tonics, and the Forge are all below, and every gallery deeper is more dangerous and richer. The Sand Worm patrols the far rim so that no one wanders off the map.

Layout

Zone	What's there	Opens with
The Rim (start pocket at the station's surface hatch)	A vast plain beside the abyss; sea pens, sand eels, the worm's tunnels on the horizon; only the Gannet rack and Juggernaut	—
The Descent	A chimney down from the hatch; ledges with Quick Brine, Needler Mk I, a workbench; the first overlook onto the abyss	750
The Upper Galleries	Glass sponge reefs and bamboo coral; Carbine, Flechette, Speed Brine; Locker spot 1; the Remnant's upper airlock	1,000
The Warrens	A maze of tubes and vents; yeti crabs and tube worms; Double Shot, Long Speargun; two overlooks	1,250

Zone	What's there	Opens with
The Station: Mess and Quarters	Flooded modules; specimen kelp tanks; Bolt Harpoon, Kick Brine; dossier terminals with the researchers' logs	1,500
The Station: Specimen Labs	Tanks (some broken: the coelacanths and salamanders are loose); Mule Brine, Deadeye; the Pale Caiman's corridor	1,750
The Station: Reactor	Power (the reactor); the Pressure Forge; the Station Chief's mech bay	2,000
The Lowest Vault	A cathedral cave open to the abyss on one side; black coral; Locker spot 2, Widow's Ink, Cold Coffin; the Lantern Leviathan's home; the best loot: two extra racks (Stormlock, Cannon Harpoon)	2,500

Overlooks: every level has at least one gallery open to the abyss. The void is a visual: a black column with drifting motes and a faint pale light far below; a diver who swims out over it sinks and dies (a 5 s pull with a warning). Traps: a pressure door (crushes a corridor), the reactor vent (a scalding jet), the vault's collapsing ledge over the abyss. Hidden quest: assemble the station's final log from five terminals to learn what the researchers were breeding.

Level mechanics

- **Down is better.** Racks and tonics deepen with the galleries; the Forge is at the reactor; the Vault's racks and Locker are the best on any map. Blood decays slowest here (the water barely moves), so every deep fight lingers.
- **The Sand Worm.** A size-7 behemoth seen tunneling on the rim's horizon. Beyond a marked line of bone stakes, the sand shakes; 8 s later, the worm surfaces and consumes the diver (no down, no revive). It never enters the caves and takes part in no web; it is the edge of the world.
- **Overlooks** are where the abyss's own beasts arrive: the colossal squid, the sperm whale, the Relict. Fighting there is fighting with your back to a fall.

- **The station's lures.** Researchers left lure beacons; the Remnant switches them on to pull beasts onto the players. Beacons can be destroyed (loud) or re-aimed (a build part).
- **Specimens.** Broken tanks released coelacanths and lab salamanders that don't fit the web: nothing hunts them but the caiman, so they multiply until the players cull them.

Flora (12)

Flora	Type	Effect
Glass sponge reef	Structure	Sharp glass: bleeds on contact; hides small beasts; shatters loudly
Tube worm garden (vents)	Producer	The vent web's base; yeti crabs and vent shrimp feed here
Bacterial mat	Producer	Carpets the warrens; amphipods swarm it
Bamboo coral	Cover	Tall stalks in the galleries
Sea pen field (rim)	Cover	Waist-high; sand eels beneath; glows when brushed, which shows a diver's path across the plain
Xenophyophore carpet	Producer	Brittle; walking on it kicks up silt that hides scent for 5 s
Black coral thicket (vault)	Cover	Dense and dark; the leviathan's hunting ground
Specimen kelp	Luminous producer	Lab-grown, glowing; draws every luminous-curious species in the station
Pressure moss	Producer	Thin growth on the station walls; harmless
Void bloom	Hazard	A luminous flower on the overlook edges that opens toward the abyss and lures curious beasts, and divers, over the lip
Brine lily	Hazard	Grows at the brine pool in the warrens; touching it or the pool: 20 damage and 5 s of slow
Lab algae tanks	Producer	Spirulina tanks the station's small fish still live on

Beasts (40, the boss, and the Sand Worm)

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Sand Eel	Fish	1	1	Plankton	Skates, snailfish, seals	Rim burrower; a school erupts underfoot
Sand Hopper	Crustacean	1	1	Detritus	Tripod fish, rattails	Rim swarm
Sea Pig	Echinoderm	1	Scavenger	Corpses, mud	Nothing	Herds on the rim; cleans corpses; free scrip, but every dead one stops cleanup
Brittle Star	Echinoderm	1	Scavenger	Detritus	Rattails	Carpets the galleries
Bone Worm (Osedax)	Worm	1	Scavenger	Bones	Nothing	Colonizes big corpses; the only thing that finishes a whale
Vent Shrimp	Crustacean	1	1	Bacterial mat	Squat lobsters, chimaera	Vent swarms
Hatchetfish	Fish	1	1	Plankton	Viperfish, dragonfish	Luminous school in the chimney; its lights follow divers' lamps
Hadal Amphipod	Crustacean	1	Scavenger (swarm)	Corpses, mats	Snailfish	A swarm that strips a corpse in 20 s and nibbles downed divers (2 per second)
Cookiecutter Shark	Fish	1	Parasite	Plugs of flesh	Nothing	Takes a round bite from beasts and divers (15) and leaves; each bite is a bleed source

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Tripod Fish	Fish	2	2	Sand hoppers, plankton	Skates	Stands on fin stilts on the rim; faces the current, showing where blood will drift
Snailfish	Fish	2	2	Amphipods, sand eels	Sixgill	Deepest galleries; gelatinous; harmless
Rattail	Fish	2	Scavenger	Corpses, brittle stars	Sixgill, goblin shark	The first to any corpse; the map's cleanup crew
Giant Isopod	Crustacean	2	Scavenger	Corpses	Nothing (armored)	Armored; rolls up when shot; slowly eats anything dead
Yeti Crab	Crustacean	2	1	Bacteria on its own arms, mats	Squat lobsters, chimaera	Vent colonies; harmless
Squat Lobster	Crustacean	2	2	Vent shrimp, yeti crabs	Chimaera, skates	Swarms the vents
Dumbo Octopus	Cephalopod	2	2	Amphipods, worms	Sixgill	Harmless drifter; a curiosity the dossier rewards
Barreleye	Fish	2	2	Siphonophore parts, jellies	Viperfish	Transparent head; sees up: it watches the overlooks
Viperfish	Fish	2	2	Hatchetfish, barreleyes	Anglerfish, chimaera	Chimney ambusher with a light lure
Fangtooth	Fish	2	2	Hatchetfish, shrimp	Anglerfish	Bites 20 and hangs on
Dragonfish	Fish	2	2	Hatchetfish	Anglerfish	A red light only it can see: it hunts divers' silhouettes against specimen kelp

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Deep Jelly	Cnidarian	2	1	Plankton	Hadal turtle	Drifts the galleries; stings (10)
Abyssal Sea Snake	Reptile	2	2	Snailfish, sand eels	Sixgill	Surfaces at the station's air pockets; venom 30 over 10 s
Siphonophore	Cnidarian	3	2	Small fish	Hadal turtle	A 20 m chain of stinging bells in the overlooks; passing through costs 40
Lab Salamander	Amphibian	3	3	Lab fish, shrimp, divers' hands	Pale caiman	Escaped specimens: bite and hold, regenerate, breed in the flooded quarters unless culled
Chimaera	Fish	3	3	Squat lobsters, shrimp, crabs	Sixgill	Ghost shark of the warrens; a venomous dorsal spine
Frilled Shark	Fish	3	3	Squid, small fish	Sixgill, Relict	An eel-like lunge; 300 needle teeth: 50 and bleed
Anglerfish	Fish	3	3	Viperfish, fangtooth, dragonfish	Goblin shark	Lure that mimics a drop bottle; a swallow pins a diver 3 s
Pelican Eel	Fish	3	3	Hatchetfish, shrimp	Sixgill	A swallow that engulfs size 2 and takes 60 from a diver
Deep Skate	Fish	3	3	Sand eels, tripod fish	Sixgill, Relict	Rim glider; harmless unless stepped on

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Coelacanth	Fish	3	3	Lab fish, squid	Pale caiman	Escaped specimen; slow, armored, ancient; breeds in the labs
Giant Sea Spider	Crustacean-like	3	2	Sea pens, jellies	Nothing	Two meters across on the rim; harmless; frightening
Oarfish	Fish	4	1	Krill, amphipods	Sperm whale	A 10 m ribbon that rises through the chimney; harmless; an omen: its appearance precedes a Predator Hunt by 60 s
Goblin Shark	Fish	4	4	Anglerfish, squid, rattails	Sixgill, Relict	A slingshot jaw from 1 m; threshold 60 blood
Elephant Seal	Mammal	4	3	Squid, snailfish	Sixgill, Relict, sperm whale (competition)	Dives from the station's air pockets to the galleries; territorial bull: 80 damage rams
Hadal Turtle	Reptile	4	1	Jellies, siphonophores	Relict	Ancient, armored, gentle; eats the overlooks' siphonophores
Pale Caiman	Reptile	4	4	Salamanders, coelacanths, researchers, divers	Relict	Bred in the labs and loose in the station's corridors; amphibious; blind, hunts by vibration

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
Sixgill Shark	Fish	5	4	Sharks, seals, chimaeras, enemies, divers	Relict	Slow, huge, patient; threshold 120; follows the amphipod swarms to corpses
Colossal Squid	Cephalopod	5	4	Sperm whale calves, sharks, divers	Sperm whale	Comes to the overlooks from the abyss; hooks grab a diver and pull toward the edge
Sperm Whale	Mammal	5	4	Colossal squid, giant squid	Nothing	Dives past the overlooks hunting squid; clicks so loud they stun small beasts; harmless to divers unless shot; a corpse feeds the whole map for a tide
The Relict	Reptile	5	4	Everything: sharks, seals, squid, the Station Chief's mech	Nothing	A mosasaur that survived down here: the map's apex, threshold 200 blood, comes up from the abyss to the overlooks and the Vault; four flippers, a crocodile's mouth on a whale's body

Beast	Class	Size	Tier	Eats	Eaten by	Behavior and note
The Lantern Leviathan	Fish	6	5	Anything in the Vault	—	An anglerfish the size of the station's reactor; its lure looks like the station's emergency light; it swallows a diver whole (teammates have 5 s); from tide 12 its lure appears in the galleries
The Sand Worm	Worm	7	Outside the web	Whatever crosses the bone stakes	—	The largest thing in the game; seen tunneling on the rim's horizon; consumes any diver beyond the stakes; no interaction otherwise

The web in motion

- **Corpses are the deep's economy.** Rattails, isopods, amphipods, sea pigs, and bone worms finish everything; the sixgill follows them. Killing the scavengers to farm scrip on the rim turns the rim into a shark lane by tide 8.
- **Lights:** hatchetfish, specimen kelp, void blooms, the anglerfish's lure, the leviathan's lure, and the divers' own lamps. Turning lamps off lowers curiosity-driven approaches by half and doubles what a diver can't see.
- **The specimens:** salamanders and coelacanths breed if ignored and only the caiman eats them; a lab overrun with specimens is loud and full of blood every time a Remnant squad passes through.
- **The overlooks:** whale and squid fight each other; the Relict eats the winner; standing on an overlook during that is standing in the blood of three apex beasts.
- **The Chief's mech** is size 5 and the Relict attacks it on sight: the Remnant's final Hunt can be ended by leading the mech to an overlook.

Definition of Done

- The rim is at least 800 m across with the worm patrolling beyond the stakes and consuming a diver on cue; the void kills correctly with warning.
- `--eco-sim void 30` shows the scavenger economy, the specimen breeding curve, and the Relict's arrival at an overlook after a whale kill.
- A four-player test reaches the Vault, sees the leviathan's lure in the galleries at tide 12, and survives a Remnant Hunt with the mech.

Presentation and technology

Red Tide is first-person 3D in raylib, extending the code that already draws the parkour side's 3D Abyss (`abyss.cpp`), with every mesh, texture, and animation generated in code, per Depth's no-assets rule. The look is inked low-poly: flat-shaded meshes with a screen-space ink outline and hatching, so a screenshot still reads as Depth.

Art

- **Style.** Low-poly meshes with flat shading, a post-process ink outline (depth and normal edge detection, line weight by distance), Bayer stipple in shadow, and the same paper-tone palette rules as the salon. Underwater fog by depth, volumetric light shafts, caustics projected on surfaces near the surface, and particle marine snow everywhere.
- **Creatures are procedural meshes.** A `CreatureBuilder` makes a body from a lathed profile curve, adds fins, limbs, shells, tentacles, spines, and eyes from parameter sets, and generates a 3-band material (base, belly, accent) from the species record. Forty species per map is forty parameter sets plus a dozen body plans (fusiform fish, flat fish, eel, ray, shark, turtle, crocodilian, crab, shrimp, cephalopod, jelly, worm, amphibian, seal, whale).
- **Animation.** Fish swim by a traveling sine along the spine; rays flap; eels undulate; crabs walk on inverse-kinematics legs; cephalopods run Chains on every arm; reptiles and mammals use a four-limb gait. Attack, flee, feed, and death clips per body plan; blood as ribbon particles that follow the scent grid's flow so what the player sees is what the web smells.
- **Weapons and hands.** First-person hands and guns are procedural meshes with the reload clips from the Weapons section; the sea-glass Forge finish is a material swap.

- **Maps are built in code from layout data.** Each map is a text layout of rooms, corridors, and props with decoration rules (which flora grows on which surface, where habitats go). Atlantis has a city generator: districts on a hill, streets by slope, building types by district, decorated by the habitat rules. The Sunken Ship is a lathed hull with decks cut from it. Caves are marched from noise fields with hand-placed anchors for the galleries and slipstreams.
- **Blood and water.** A red tint at the mask edge that grows with local scent; visible blood plumes; the scent meter and predator pulse on the HUD.

Performance budget

Item	Budget
Active agents near players	300 at 10 Hz decisions, 60 Hz motion
Population counters beyond 120 m	Unlimited (no bodies)
Mesh instancing	One draw per species per LOD
LODs	3 per creature; billboards past 80 m
Scent grid	2 m cells, updated at 1 Hz, in a background thread
Target	60 fps at 1080p on a mid-range laptop with four players

Networking

- Host-authoritative through the shared arcade layer (GameNetworkingSockets). The host runs the ecosystem and hit validation.
- **Client prediction** for the local diver's movement; **lag compensation** for hits: the host rewinds beast positions up to 200 ms to judge a shot, so aiming feels local.
- Snapshots at 20 Hz with delta compression; beasts and enemies interpolated 100 ms behind; interest management by distance (200 m) and line of sight.
- Scent and alarm live only on the host; clients receive the HUD values for their own position.
- Disconnect: the diver is downed and held for 2 minutes; reconnecting restores them if the body is revived.

Tools

- `--eco-sim <map> <minutes>` (headless ecosystem), `--redtide-shots` (every creature, weapon, and map zone rendered to PNG), `--redtide-sim <map> <tides>` (a bot team plays to a tide count and reports scrip, downs, and which beasts did the killing), `--net-loop redtide`.

Sound design

Red Tide’s sound is the ecosystem made audible: every species has a voice, every kill has a wet consequence, and the music is the round timer. All of it is synthesized through the cue registry and buses defined in the Depth master reference.

Underwater mix rules

- Everything is low-passed and reverberant; gunfire is a thump and a hiss of bubbles, not a crack, except the powder guns, which crack and then boom in the water.
- Distance is muffling: sounds beyond 40 m lose their highs entirely, so the direction of a far predator is felt more than heard.
- The diver’s own breathing is always present and quickens with low HP and nearby blood.

Music

State	Music
Calm between tides	A slow brass-and-glass drone in the map’s key; the 20 s countdown ticks under it
Tide (early)	A pulse and a bass line; adds a melodic layer when the quota is more than half done
Tide (late)	Driving percussion, distorted brass; a siren layer at 20+
Blood rising	A low string swell that grows with the local scent value; the strings are the scent meter you can hear
Predator approaching	The map’s apex motif enters, panned from its direction

State	Music
Hunt	A faction theme: the Wreckers' shanty on broken accordion, the Drowned's bell and choir, the Raiders' drums, the Lost Ones' organ, the Remnant's alarm klaxon and clockwork
Boss present	The boss theme; everything else ducks
Downed	Music thins to a heartbeat and the breathing
Match over	A slow shanty over the score

Each map has its own palette: the ship's creaking brass, the cave's echoes and drips, the reef's bright marimba-like synths, Atlantis's choir and organ, the Void's sub-bass and near-silence.

Ambience

Map	Bed
The Sunken Ship	Hull groans, the list's creak, sprats fluttering, the air pocket's slosh, the crocodile's breath from the salon
The Underwater Cave	Drips, slipstream rush, cricket chirps in the dry chambers, the swiftlets, distant rockfall
The Coral Reef	Snapping shrimp crackle, parrotfish crunch, surf above the lagoon, dolphin whistles, whale song at the wall
Atlantis	Aqueduct flow, the cisterns' echo, snapping shrimp in the walls, cormorant wings, the chapel's god-pool
Approaching the Void	Almost nothing: the reactor hum, vent hiss, sperm whale clicks, the worm's distant rumble, the void's low tone

Effects

- **Every beast** has a voice cue (clicks, grunts, croaks, whistles, or silence by design), a pain cue, a death cue, and a feeding cue (tearing, crunching, slurping). Feeding cues are loud by design: they tell the players where the corpses are being eaten and how many things are eating.
- **Blood** has a sound: a wet bloom on a kill, a spreading hiss as a plume grows.
- **Weapons:** fire, reload steps, empty click, the Forge's crush and hiss, the Locker's eel snap and the chest sinking, the Harpoon Cannon's boom.
- **Tonics:** each machine has a jingle (the Zombies tradition), a bottle clink, and a gulp.
- **Drops:** a bell for a drop appearing, a distinct sting per drop.
- **The alarm:** no sound tells the player the alarm value, but the faction's arrival has a tell: the Wreckers' cutters scraping the hull, the Drowned's bell, the Raiders' drum, the Lost Ones' marching, the Remnant's airlock cycling.
- **Tells for hazards:** the sand worm's rumble, the Reacher coral's creak, the bobbitt worm's hiss, the toad's croak, the Wyrms' grate rattle, the leviathan's lure hum.
- **HUD and UI:** the tide bell, the quota tick, the scrip counter's coin, the down and revive stings.

Accessibility

Every audio tell has a visual: a directional pulse on the mask, a caption for barks, and a scent meter for the blood strings.

Balance, simulation, and build order

Red Tide is balanced against Zombies' pacing (a competent team reaches tide 20 in about 40 minutes) and against its own ecosystem, which must punish careless killing without making careful play boring.

Balance targets

Measure	Target
Tide reached by a bot team of four playing carefully	25 ± 3 on the Sunken Ship, 20 on Atlantis and the Void

Measure	Target
Tide reached by a bot team that shoots everything	12 ± 3 (the ecosystem must end them)
Time to tide 10	15–20 minutes
First apex arrival after 100 blood in one region	30–90 s by map
Share of deaths caused by beasts, enemies, hazards	About 60 / 25 / 15 at tide 15
Scrip at tide 10 for a careful player	15,000–20,000 (enough for Juggernaut, two tonics, a rack gun, and one Forge)
Enemy squads per tide from the alarm, careful team	0.5 · careless team: 3
Weapon cost-efficiency	Damage per second per 1,000 scrip within ±20% across rack weapons of the same tier; Locker weapons 1.5×; Forged 2.5×

Simulation tools

- `--eco-sim <map> <minutes> [pattern]` runs the ecosystem with a scripted player pattern (quiet, loud, camping, spread) and prints population curves, blood levels by region, and predator arrival times.
- `--redtide-sim <map> <tides> [careful|careless]` plays a bot team and reports the balance targets above.
- `--web-check <map>` asserts every species has at least one predator or a reason not to (toxic, apex, boss), every tier-1 species has a producer, no diet cycle is unreachable, and every corpse has a scavenger.
- All numbers live in `redtide_species.json`, `redtide_weapons.json`, and `redtide_maps/`; nothing is inline.

Build order

Stage	Work	Gate
1	Ecosystem engine headless: species records, scent grid, sound layer, state machine, population; twelve generic species	<code>--eco-sim</code> shows the sprat chain end to end
2	3D core: inked low-poly renderer, <code>CreatureBuilder</code> with the body plans, first-person diver, the Cormorant	A diver swims a box room among generated fish
3	The Sunken Ship: layout, all 40 beasts and 12 flora, racks, tonics, the Locker, the Forge, drops, tides, the Wreckers, the Goliath	Sunken Ship Definition of Done
4	Multiplayer through the shared arcade layer: prediction, lag compensation, snapshots, downs and revives over LAN	Four players finish a Sunken Ship match on LAN
5	The Underwater Cave (slipstreams, dry chambers, rockfall)	Its Definition of Done
6	The Coral Reef (Reacher coral, dolphins, the flats)	Its Definition of Done
7	Atlantis (the city generator, aqueduct blood flow, cisterns)	Its Definition of Done
8	Approaching the Void (the rim, the worm, the station, the Vault)	Its Definition of Done
9	Charms, skins, dossiers, hidden quests, sound pass	<code>--audio-test</code> and the dossier complete for every species
10	Internet play through the relay server; balance pass with <code>--reddtide-sim</code> on every map	All balance targets met or logged

Open questions for the designer

- ☐ Should killing a dolphin or a whale carry a scrip penalty on top of the ecological one?

- ☐ Should the boss beasts have a chance to appear in earlier tides as a rare event, or stay strictly tied to tide 12 and Predator Hunts?
- ☐ Should there be a sixth map later, or should new content go into the existing five as "seasons" of species?

Map data workbooks

Every number a map needs lives in one workbook per map, which Claude Code exports to `redtide_maps/<map>/*.json`. The Sunken Ship workbook (`RedTide_SunkenShip_data.xlsx`) is the template; the other four maps get the same sheets.

Sheet	Holds
Species	One row per beast: size, tier, tags, archetype, social group, home zone and flora, defend radius, blood threshold, temperament (aggression, fear, curiosity), senses (sight, scent, hearing, electroreception), weak point, front armor, base HP and bounty, blood on death and per second, speed and turn rate, drop chance
Attacks	One row per attack move: damage, windup, cooldown, range, the visible tell, and the effect (bleed, hold, knockback, poison, stun)
Diet	The 40 × 49 preference matrix: each predator's row sums to 1 across the species it eats plus plankton, corpses, parasites, each flora, and the enemy faction
Flora	Type, zones, effect, contact damage, grazers, regrowth
Spawn	Per zone: species, count at tide 1, respawn seconds, capacity growth
TideCurve	Tunables and the per-tide table: quota, HP and bounty multipliers, beast damage multiplier, alarm threshold, enemy spawn chance, blood decay, Hunt schedule, apex wander, boss availability, tide bonus

Sheet	Holds
Blockout	Zones with coordinates and door costs, links, points of interest, training loops, blood-flow direction, sightlines, alarm regions
Boss	The map boss's phases, arena rules, and rewards

The tide curve (from the TideCurve sheet)

Value	Rule	Tide 1	Tide 5	Tide 10	Tide 20
Kill quota (4 players)	$12 \times 1.15^{(\text{tide}-1)}$	12	21	42	170
Beast HP	$\times 1.08$ per tide	1.0	1.36	2.0	4.3
Bounty	$\times 1.10$ per tide	1.0	1.46	2.36	6.1
Beast damage	$\times 1.05$ per tide, capped at 2.5	1.0	1.22	1.55	2.5
Alarm threshold	$100 / (1 + 0.1 \times \text{tide})$	91	67	50	33
Enemy spawn chance per roll	$10\% + 3\% \times \text{tide}$ from tide 4, cap 70%	0	25%	40%	70%
Blood decay per second	2% / 1% / 0.5% by era	2%	2%	1%	0.5%
Hunt	Every 5th tide; Predator Hunts possible from 10	—	Faction	Either	Either
Apex wander	From tide 10	no	no	yes	yes
Boss may appear	From tide 12 (Predator Hunts from 10)	no	no	Hunt only	yes

Quota scales 0.4 / 0.6 / 0.8 / 1.0 with 1–4 players. All of these are single cells in the Tunables block, so a balance pass edits one sheet.

Enemy factions in detail: the Wreckers (template)

Each faction gets this treatment; the Wreckers are written out first because they belong to the first map. Enemy HP rises 8% per tide, weapons upgrade every 5 tides, and every unit has a tell and a bark set.

Units

Unit	HP	Weapon	Damage	Speed	Role	Tell	Loot
Cutter	120	Boarding knife	25 melee, 0.8 s	Fast (2.4)	Flanker; goes for downed divers and lone divers	Bubbles from a fast exhale before a lunge	200 scrip
Speargunner	150	Long Speargun	60, 1 shot per 3 s	Diver speed	Covers the squad from 15–20 m; aims for the diver reviving	A red laser dot from the speargun's sight, 1 s before firing	Long Speargun, 250
Netman	180	Net Gun and knife	Net holds 4 s; knife 20	Diver speed	Nets a diver, then the Cutters close	Swings the net overhead twice	Net Gun, 250
Chummer	140	Chumthrower	0	Diver speed	Lobs chum at divers' feet to call the reef onto them; retreats	Shakes the chum bucket	Chumthrower, 300
Foreman (Hunts only)	900	Harpoon Cannon	200 per bolt, 1 per 4 s	Slow (1.4)	Anchors the Hunt from the hull breach; shouts orders that buff Cutters +20% speed	Winds the cannon's winch for 2 s	Harpoon Cannon (10 bolts), 600

Squad tactics

1. A squad enters at the hull breach and moves Stern → Engine → Cabins → Salon, checking each room's corners; it will not enter a room with a size-4 beast in it.
2. On contact: the Netman targets the closest diver, the Speargunner takes a firing position 15 m back, the Cutters flank through the second doorway, the Chummer throws chum behind the divers so the reef arrives from the direction they retreat to.

3. A Wrecker below 30% HP retreats toward the breach and calls for help (+1 Cutter if the alarm is still above threshold).
4. Wreckers ignore beasts below size 3 and fight anything bigger that attacks them; they will abandon a fight with divers to flee a shark.
5. In a Hunt, the Foreman holds the Stern; the squad's job is to drive the divers toward him.

Barks

Spoken as gurgling radio chatter (formant synth, subtitled). Each unit has 6–8 lines per situation; examples:

- Spotting divers: "Divers on the wreck. Nobody's collecting our salvage." / "Four of 'em. Chum 'em."
- Chum thrown: "Dinner's served, fish." / "Let the reef do the work."
- Shark arrives: "That's not ours! Back to the breach!" / "Cut the chum, cut the chum!"
- A Wrecker dies: "Marlow's gone. Keep working." / "The safe's worth it. Push."
- Netting a diver: "Got one wrapped!" / "Pull it in, pull it in."
- Foreman: "The safe's mine, and so's the ship." / "Winch up. Put a bolt through the helmet."

The same treatment for the other factions

The Drowned (Deckhand, Lantern Bearer, Bosun), the Reef Raiders (Spear Diver, Turtle Rider, Sling Diver, Shaman), the Lost Ones (Legionnaire, Cultist, Slinger, Armored Lost One, Priest), and the Remnant (Sentinel, Researcher, Cultist, Station Chief) each get a units table, five tactics, and bark sets in their map's workbook on a Faction sheet.

Boss fights

Every map boss follows one template: three phases at 100–60, 60–30, and 30–0% HP; an arena that is a room until tide 12 and the whole map after; one way to blind, stun, or expose it that rewards knowing the map; and a reward of 1,500 base scrip, a Locker weapon per diver, and a quest key.

The Goliath (the Sunken Ship)

- **Arena.** The Engine Room until tide 12; after that it wanders and the ship is the arena. 15,000 HP; front plates halve damage; the gills behind the plates are the weak point, exposed for 2 s after an Inhale and after a Tail Slam.
- **Phase 1 (100–60%).** Inhale (a 1.5 s windup as the gills close and water rushes; pulls one diver from 4 m into its mouth; the team has 5 s to deal 150 damage to the gills or the diver is swallowed and downed), Lunge (140, bleed), Boom (a 10-noise pulse that stuns 1 s in 10 m). It stays within 12 m of its home.
- **Phase 2 (60–30%).** Adds Tail Slam (90, knocks back 6 m, drops loose cargo on three marked squares), Boom every 3 s, and calls two Groupers. It roams the whole room; the Forge can't be used while it is within 8 m.
- **Phase 3 (30–0%).** Enraged: speed $\times 1.5$, Inhale every 8 s, a swallow is an instant down. It breaks the aft bulkhead and chases into the Stern.
- **Arena rules.** Shooting the luminous algae mat off the wall (300 damage) blinds it for 6 s because it hunts by the mat's light; cycling the power switch stuns it 2 s (once per fight); loose cargo can be shot to drop on it for 200.
- **Team plan.** One diver baits the Inhale from 4 m; two shoot the gills in the 2 s window; one handles the Groupers and the cargo. Reward: 1,500 base scrip $\times$ the tide multiplier, a Locker weapon each, and the captain's safe key. It reforms in its home after 6 minutes at 50% HP.

The other four, in brief (full sheets in each map's workbook)

Boss	Arena	The expose	Phase 3 twist
The Lobster (Cave)	The Cathedral; from tide 12 any hall via slipstreams	Ringling the crystal coral makes it rear up and show its underside for 3 s	Drops the Cathedral's stalactites on a timer; crushes a Shell Shield outright
The Matriarch (Reef)	The Drop-off Wall and the Blue Hole; from tide 12 the flats and the lagoon	Killing her two pod members makes her charge in straight lines; the Bommie's surge channel flips her for 4 s	The pod herds divers off cover; she rams the Bommie and collapses the cleaning station

Boss	Arena	The expose	Phase 3 twist
The Cistern Wyrms (Atlantis)	The chapel's god-pool; from tide 12 every drain, so the city is the arena	The god-pool trap drains and strands it for 6 s; netting three grates forces it to surface at the fourth	It hunts the streets and drags divers into the cisterns; the aqueduct fills with its blood and calls the great white onto it
The Lantern Leviathan (Void)	The Lowest Vault; from tide 12 its lure appears in the galleries	Shooting the lure (500) blinds it 5 s; the reactor vent scalds it for 10%	Its lure splits into three; the ledge collapses; the Relict may arrive to fight it for the corpses

HUD and UI

The HUD is drawn as the inside of a brass diving helmet: gauges at the edges, the water in the middle. There is no tutorial; the how-to guide in the arcade menu and the dossier are the only teaching, so every gauge must be readable at a glance.

In-match HUD

Element	Where	What it shows
Health	Bottom left, a brass pressure gauge	0–250; the needle shakes when hit; the glass cracks below 25
Scrip	Top right, a chalk tally on the visor	Current scrip; +10 and bounty numbers float up from kills in ink
Tide and quota	Top center, a tide dial	Tide number; a ring that fills toward the quota; the 20 s calm countdown between tides
Weapon and ammo	Bottom right	Weapon name, magazine and reserve as brass counters, the Forge finish shown as sea-glass on the counter, the alternate-ammo glyph

Element	Where	What it shows
Second weapon and tactical	Under the ammo	Small icons; the tactical count
Tonics	Bottom center, bottle icons	Up to five bottles; a cracked bottle after a down
Scent meter	Left edge, a thin vertical vial	Fills red with local blood; the visor's edge tints red as it fills
Predator pulse	Around the visor	A dull pulse from the direction of any approaching beast over size 3 within 40 m; its rate rises with proximity
Alarm	No direct display	The faction's arrival tell (sound and a visor flicker) is the only warning, by design
Teammates	Top left	Name, health bar, a down icon with a 45 s ring, distance and an arrow when off-screen
Prompts	Center bottom	Contextual: buy (name and cost), revive, build, pick up, refill; drawn as chalk on the glass
Drops	Center, brief	The drop's name and 30 s timer
Downed	Whole visor	Water rises inside the helmet; the Cormorant only; a heartbeat
Hit markers	Center	A small ink splash; gold for weak points; a crack sound for kills
Captions	Bottom	Barks and radio chatter, toggleable

Menus

- **Arcade drum reel:** the game's attract loop is the Sunken Ship at night with the Goliath's lamp.
- **Lobby:** 1–4 slots, map picker with a tide record per map, mode picker (see Modes), skin picker, charm pouch (5 slots), ready lever.

- **How to play:** one scrolling page in the ship's log style: the loop, scrip, blood, the alarm, downs, the Locker, the Forge, tonics, drops. Six short paragraphs and one diagram of the blood rules. Nothing else teaches.
- **Dossier:** a tabbed bestiary per map; pages unlock by kill or 30 s of observation; each page shows the beast's silhouette, class, diet, predators, weak point, and its dossier note.
- **Records:** per map, highest tide, most scrip, fastest Forge, longest survival, and team records.
- **Pause** (host only, 3 per match): the salon behind the porthole, the team list, options.
- **Results:** the tide reached, scrip per diver, kills by class, deaths by cause, beasts eaten by beasts because of you, arcade tokens earned; a slow shanty.

Controls (keyboard and mouse; gamepad mapped the same)

Move, swim up and down (space and ctrl), sprint (shift, with a stamina bar that only shows while sprinting), aim, fire, reload, melee (V), tactical (G), swap weapon (1, 2, 3), interact (E, hold for purchases over 1,000), lamp on and off (L), ping (middle mouse: a marker teammates see, with a voice line), charm (T), scoreboard (Tab).

The divers and their quips

The four divers are four suits, not four characters: the Diver, the Whaler, the Stowaway, and the Mechanic, each with a fixed silhouette and a voice, and each replaceable by a skin. Their lines are about the situation, not their lives, so any skin can say them; the fun is that they talk to each other.

How quips work

- Each diver has a voice (a formant synth pitch and rhythm) and a line set per situation. Lines are picked from the set, never repeated within 3 minutes, and only one diver speaks at a time.
- **Callbacks:** about a third of lines are two-part: one diver speaks, and a teammate within 15 m answers 1–2 s later with a reply from their own set. The four voices differ in tone: the Diver is dry, the Whaler is gruff, the Stowaway is cheerful and slightly drunk, the Mechanic is nervous and precise.
- Quips are subtitled and never interrupt a tell or a Hunt announcement.

Situations (each with 6–10 lines per diver)

Situation	Example exchange
Match start	Stowaway: "Who brought the snacks?" Mechanic: "We're the snacks."
First blood in the water	Diver: "That's blood. Watch the doors." Whaler: "Let it drift. Don't feed it."
Scent meter high	Mechanic: "Meter's red. Very red." Stowaway: "Ooh, red's my favorite."
Predator pulse	Whaler: "Something big's coming from the stern." Diver: "Then we go to the bow."
Someone shoots a harmless big thing (dugong, whale, sturgeon)	Diver: "Why would you shoot that." Stowaway: "It looked at me funny."
Killing a cleaner	Mechanic: "That was a cleaner. The groupers are going to be in a mood."
Chum thrown at the team	Whaler: "They're chumming us! Move!"
Downed	Stowaway: "Bit of a lie-down, don't mind me." Whaler: "Get up, we're not carrying you."
Revive	Diver: "You're welcome. Again." Mechanic: "Please stop doing that."
Buying a tonic	Stowaway: "Bottoms up." Diver: "That's your fourth."
Locker pull, bad	Mechanic: "A Gannet. From a chest. Wonderful."
Locker pull, wonder weapon	Whaler: "Now THAT'S a fish-killer."
Forge	Diver: "Pressed and ready." Stowaway: "Shiny."
Hunt begins	Whaler: "Wreckers! Cover the breach!" Mechanic: "Or let the crocodile handle it?"
Enemy eaten by a beast	Stowaway: "Ha! Reef's on our side." Diver: "For now."

Situation	Example exchange
Boss appears	Mechanic: "That's... that's a fish. That's a very large fish."
Boss swallow	Diver, from inside: "It's warm in here. Shoot the gills!"
Tide cleared	Whaler: "Breathe. Buy. Go again."
Quiet tide (no gunfire for 60 s)	Stowaway: "Too quiet. I don't like it. I do like it."
Last diver standing	Diver: "Fine. Just me, then."
Match over	Mechanic: "Tide twelve. We'll get thirteen." Stowaway: "Next time, more snacks."

Environmental lore (light, on purpose)

The divers work for the Owners, a salvage concern that pays scrip for cleared water and asks no questions. Each map's dossier terminals, chalk marks, and Wrecker chatter drop a line or two: the *Marguerite* went down with the captain's safe; the cave swallowed an expedition whose lanterns still burn; the reef's tribe hunts because the Owners pay them too; Atlantis's citizens drowned and kept walking; the station bred what it found in the abyss and the abyss bred back. That's all the story the game needs: the players supply the rest over voice chat.

Modes

All modes run on the same maps and ecosystem; each changes the rules, not the content. The host picks a mode in the lobby.

Mode	Rules	Why play it
Red Tide (standard)	Tides, quotas, Hunts, as designed	The main game
Blackout	Lamps off, no luminous flora, sonar pings only	Horror: the reef by ear

Mode	Rules	Why play it
Quiet Water	Powder guns and explosives removed; knives, spearguns, needlers, nets only; alarm halved; blood decays slower	The stealth game hidden inside the shooter
Feeding Frenzy	Blood never decays; every kill bleeds 3×; Blood Frenzy drops constantly; tides are 60 s each	Ten minutes of chaos
Quota	No tides: survive until a timer, with the Owners' quota rising every 2 minutes; scrip only from beasts over size 2	Rewards patience and target selection
Aquarium	No enemies, no Hunts, no tides; infinite scrip; every beast at once; a free camera	Watch the web, learn tells, test builds, take screenshots
Salvage Run	One life each, no revives, a 15-minute timer, and five salvage crates to carry to the extraction point through the whole map	A heist
The Long Night	A four-hour match saved between sessions (host-side save); tides continue where they stopped	Friends across a week
Apex	Every Hunt is a Predator Hunt; the boss is awake from tide 1; bounties doubled	For teams that have beaten tide 30
Draft	Each diver gets a different starting loadout from a shared pool, no Locker, no racks; weapons only from enemies	Forces the team to hunt Wreckers
Poachers	The four divers are split 2 v 2: each pair earns scrip from the same reef; pairs can't hurt each other but can chum each other; the richer pair at the timer wins	The only competitive mode

Mode	Rules	Why play it
Custom	Sliders for every tunable in the TideCurve sheet, plus toggles for each faction, boss, and flora hazard	Whatever the table wants

Wonder weapon quests

The Locker deals every wonder weapon, always. On the Coral Reef, Atlantis, and Approaching the Void a quest also awards it, guaranteed, to whoever finishes:

- **Reef: the Anemone Gun.** Feed the Bommie's cleaning station three parasites (brushed off teammates into a jar), survive one Raider drum without shooting a dolphin, then carry a live blue-ringed octopus (netted) to the Blue Hole's ledge without being bitten.
- **Atlantis: the Tide Staff.** Light three chapel braziers with the reactor-crystal's spark carried from the treasury, hold the plaza through one tuna crossing, and offer a Lost One's ichor (a Chummer jar filled from a corpse) at the god-pool while the Wyrms is submerged.
- **Void: the Abyssal Lure.** Assemble the station's final log from five terminals, re-aim three lure beacons toward the abyss, and lure the Relict to the reactor vent, where the scald exposes its belly for one bolt from the Cannon Harpoon.

Species seasons

Instead of a sixth map, each season adds five to ten species across the five maps that change the webs:

Season	Theme	Examples
1: Invaders	Species that don't belong and outcompete	Lionfish on the reef (eats everything small, nothing eats it); Asian carp in the Atlantis farms; a tropical crocodile in the cave's warm pool

Season	Theme	Examples
2: Migration	Passers-through that arrive only in some tides	A tuna run in the Sunken Ship's shark lane; a whale calf on the reef that the Matriarch hunts; salmon through the cave's slipstreams
3: Parasites and plagues	New parasites and a fungal flora that spreads	Isopods that replace fish tongues; a red-tide algae that blinds; a cordyceps for crabs
4: Giants	One new size-5 per map	A whale shark, a giant manta, a leatherback in the cave, a basking shark over Atlantis, a giant oarfish
5: Night	Nocturnal casts that only appear when lamps are off	Flashlight fish, vampire squid on every map, a night-shift shark

A season is a data drop: species rows, diet matrix columns, spawn entries, and dossier pages, with models from the existing body plans.

Weapon handling and the Forged list

Every weapon gets handling numbers (how it feels) and a Forged identity (what it becomes). The Forge still gives +150% damage and +50% ammo, but each weapon's Forged form has a name and one changed behavior, the way Pack-a-Punch weapons do.

Handling

Spread is in degrees at rest; ADS is aim-down-sight time in seconds; swim multiplier is speed while aiming; headshot is the weak-point multiplier for the class (Deadeye raises every class by 1.0).

Class	Spread hip / ADS	ADS time	Swim while aiming	Recoil	Headshot	Falloff
Gas pistols	2.5 / 0.8	0.20	0.9	Low, vertical	2.0	Full to 15 m, 50% at 25 m
Needle repeaters	4.0 / 1.5	0.25	0.7	Medium, rising	1.8	Full to 20 m, 50% at 30 m

Class	Spread hip / ADS	ADS time	Swim while aiming	Recoil	Headshot	Falloff
Powder carbines and rifles	2.0 / 0.5	0.30	0.6	High, kick and settle	2.2	Full to 30 m, 50% at 45 m
Scatter guns	8.0 / 6.0	0.30	0.6	High	1.5	Full to 6 m, 50% at 12 m
Spearguns and harpoons	1.0 / 0.2	0.45	0.4	Single heavy kick	2.5	Full to 60 m; bolts drop after 40 m
Needle LMG	5.0 / 2.5	0.60	0.4	Medium, constant	1.6	Full to 20 m
Launchers	3.0 / 1.0	0.50	0.5	Heavy	none	Charges sink after 25 m
Melee	—	—	1.0	—	2.0	1.5 m, 2.5 with the Trident

Every hit has a hit marker; weak-point hits ring; a bolt that pins a size-1 or size-2 beast to a wall holds it for 4 s and can be recovered.

The Forged list

Weapon	Forged name	The twist
The Cormorant	The Albatross	3-dart burst; the Last Stand pistol keeps this form
Gannet	Frigatebird	Darts ricochet once off hard surfaces
Twin Gannets	Storm Petrels	Alternating fire never needs to reload both at once (one reloads while the other fires)
Needler Mk I	The Hornet	Darts leave a 2 s bleed
Needler Mk II	The Wasp's Nest	Every 10th dart is a tracer that marks the target for the team (Chum Rounds effect, 5 s)
Gatling Needler	The Swarm	No spin-up; magazine 200
Reef Rattler	The Triple Reef	4-round burst; the fourth is a weak-point seeker within 5 degrees

Weapon	Forged name	The twist
Sea-Pattern Carbine	The Lieutenant	Headshots refund 2 rounds
Trawlerman's Rifle	The Skipper	Penetrates two beasts in a line
Stormlock	The Typhoon	Full auto with no recoil climb for the first 10 rounds of a magazine
Flechette 12	Twelve Apostles	Flechettes stick and detonate 1 s later (small, silent)
Drum Flechette	The Hurdy-Gurdy	Fires 2 shells per trigger pull
Long Speargun	The Quiet Word	Kills of size 1-2 make no blood
Bolt Harpoon	The Judgement	Bolts pin size 3 for 3 s and size 4 for 1 s
Cannon Harpoon	The Cathedral Gun	Bolts explode on any size and chain-pull a size-3 corpse to you (a lure you can carry)
Limpet Launcher	The Barnacle King	Charges stick to beasts and detonate on the next hit instead of a timer
Net Gun	The Fisherman's Friend	Nets hold size 4 for 4 s and size 5 for 2 s
Chumthrower	The Dinner Bell	Chum spots also apply Weaken (–20% attack) to beasts feeding there
Diver's knife	The Filleter	Kills of size 1-2 are instant from behind
Boarding Axe	The Boarder	Sprint-swing knocks back 3 m
Tesla Gaff	The Lightning Rod	Chains to 4 and stuns 2 s
Trident	Neptune's Fork	Throwable, returns in 3 s
Sawtooth	The Shipwright's Saw	Bleed lasts 10 s and doubles blood: a weapon that is also bait
The Galvanic Rod	The Lightning Keel	Arcs leave a 5 s electric field
The Resonator	The Deep Bell	Its ring cancels blood scent in the cone for 5 s
The Anemone Gun	The Garden	Polyps poison and spread once

Weapon	Forged name	The twist
The Tide Staff	The Sovereign's Staff	The wave drowns enemies in air pockets and carries blood with it
The Abyssal Lure	The Deep Call	The burst calls one apex to the point on your side for 30 s

Re-rolling the alternate ammunition costs 2,500; Forging a Locker weapon also re-arms it. All Forged weapons carry the sea-glass finish under any skin.

Enemy spawning and Hunts

Alarm squads

1. Each region tracks its alarm (constants in the engine workbook). Above threshold, a roll every 5 s spawns a squad with the tide's chance; the alarm halves when it succeeds.
2. The squad spawns at the region's nearest **faction entry** (each map lists them) that is at least 35 m from every diver and out of every diver's sight; if none qualifies, at the map's default entry.
3. Squad size is 3 at tides 4–9, 4 at 10–19, 5 at 20+, drawn from the faction's composition (never two of the specialist unit in one squad).
4. Squads path toward the region where the alarm fired, then toward the nearest diver's last known position; they lose the trail after 20 s without sight, scent, or sound, and then patrol the faction's route.
5. At most 2 alarm squads are alive at once (3 at tide 20+); further rolls are skipped.
6. A squad wiped by beasts counts as resolved: the alarm is not raised by their deaths.

Hunts

- Tide 5, 10, 15, and every 5th tide after: a **Faction Hunt** unless a Predator Hunt is rolled (50% from tide 10, 70% from tide 20).
- **Faction Hunt.** The quota is replaced by "eliminate the Hunt force": 10 units at tide 5, 12 at 10, 14 at 15, 16 at 20+, plus the faction's Hunt leader. The force spawns in three waves 30 s apart from the map's Hunt staging point; the leader spawns with the third wave.

The Hunt ends when the force and the leader are dead or eaten. Reward: a Resupply drop and a guaranteed Locker weapon for each diver (a free pull from the nearest spot).

- **Predator Hunt.** The quota is a fixed number of apex kills: 2 at tide 10, 3 at 15, 4 at 20+ (the map's boss counts as 2). The map's apex beasts are given the divers' position as a blood source of 500 and arrive within 60 s from their home regions. Reward as above.
- During any Hunt the ecosystem keeps running; beasts eat Hunt forces and Hunt forces fight beasts. A Hunt has no time limit but the calm between tides is doubled to 40 s after one.
- The oarfish (the Void) and the drum (the Reef) are pre-warnings of 60 s; other maps warn with the faction's tell 20 s before the first wave.

Solo and duo

- Quotas scale 0.4 / 0.6 / 0.8; squads are 2 / 3 / 3 units; Hunt forces are 5 / 7 / 9; Predator Hunt kills are 1 / 2 / 3.
- Solo: Quick Brine gives a self-revive (3 per match); a downed solo diver has 60 s.
- Duo: a downed diver has 55 s; revives take 3 s.

Hidden quests: step lists

Each map has one hidden quest. Steps are ordered; a step that fails (the item is lost, the timer runs out) resets only that step. Every quest has a fail-safe: nothing in it can make a match unwinnable.

The Sunken Ship: the captain's safe

1. **Trigger:** read the captain's log on the bridge (a chalk prompt appears after tide 3).
2. Collect three keys: one drops from the Goliath (any kill), one from the Wreckers' Foreman (a Hunt), one is held by the Common Octopus (kill it in its den; if it has stolen an item, the key is with it).
3. Carry all three to the safe on the bridge; opening takes 20 s and rings the ship's bell (10 noise). Fail: a diver holding a key who dies drops it where they fell.
4. **Reward:** 3,000 scrip each, a Forged wonder weapon (the Lightning Keel) for the opener, and the dossier page "The Owners".

The Underwater Cave: the expedition's key

1. **Trigger:** the Lobster's kill drops the expedition's key.
2. Find the three lanterns the Drowned left burning (one in each dry chamber) and put them out with an ink cap or an ink bomb; each doused lantern silences the swiftlets for 5 minutes and drops a page of the expedition's log.
3. With the key and three pages, open the Lantern Cache's locked crate in the Cricket Cavern during a tide (not the calm). The crate takes 15 s and the Drowned come for it (a mini-Hunt of 6).
4. **Reward:** a free wonder weapon (the Resonator), 2,000 scrip each, and the Drowned no longer spawn from the Lantern Cache for the rest of the match (they use the Chimney instead).

The Coral Reef: the drum

1. **Trigger:** kill a Shaman (Hunt) and pick up the drum.
2. Carry the drum to Turtle Beach's altar without beating it. A drum-carrier swims at 80% speed; beating it (a temptation the HUD offers) calls sharks and resets the step.
3. At the altar, survive one tide on the beach while the turtles nest (the altar counts nests; 5 nests complete it). Raiders target the beach.
4. **Reward:** the Anemone Gun (Forged), the dolphin pod becomes allied again if it was hostile, and hammerheads avoid Turtle Beach for the match. This is separate from the wonder-weapon quest for the Anemone Gun, which stays as written in Modes.

Atlantis: the lighthouse

1. **Trigger:** read quest terminal 1 (customs house).
2. Read terminals 2–5 (wall fort, scroll store, treasury, crypts) in any order; each reveals one of the four wall-fires' positions on the rampart.
3. Light the four wall-fires with the reactor-crystal's spark, carried from the treasury in a jar (the jar breaks if the carrier is downed). The rampart walk is 1,500 m of fire coral and barracuda.
4. Climb the lighthouse and light it: the whole city appears on the sonar for the rest of the match (every grate, every Lost One, every apex).
5. **Reward:** the Wyrms' key (opens the treasury vault: 5,000 scrip split) and the Tide Staff (Forged) for the lighter.

Approaching the Void: the final log

1. **Trigger:** read terminal 1 in the mess.
2. Read terminals 2–5 in order (bunks, dry lab, holding pens, mech bay); each requires the previous one and each spawns a Remnant squad on read.
3. Open tank 7 in the wet lab with the station's master key (the Leviathan's kill drop). The Relict egg inside is a carried item; the Relict comes for it wherever it is (a permanent blood source of 300 to the Relict only).
4. Carry the egg to Overlook 5 and place it on the collapsing ledge; when the Relict climbs onto the ledge for it, drop the ledge (the trap). The Relict falls into the abyss; it does not return this match.
5. **Reward:** the Abyssal Lure (Forged), 5,000 scrip split, and the final log's dossier page: what the station was breeding. Fail-safe: if the egg falls into the void, the Relict follows it and the quest ends without the weapon.

The wonder-weapon quests (Reef, Atlantis, Void)

The three quests in Modes stay as written and are separate from the above; both can be completed in one match.

Ecosystem test plan

`--eco-sim <map> <minutes> <pattern>` and `--web-check <map>` assert the following before any art exists. Patterns: quiet (knife only), loud (every kill with a powder gun), camping (one room, every kill), spread (four divers in four regions).

Every map

- ☐ `--web-check` : every species has a predator or a listed reason not to; every tier-1 species has a producer; every corpse has a scavenger within one region; no diet row references a missing species.
- ☐ Population is stable for 10 minutes with no divers (no species goes to 0 or above 3× its spawn count).
- ☐ A size-3 corpse in still water reaches a threshold-30 predator's cue at 20 m within 60 s and a threshold-120 apex only when four such corpses exist.
- ☐ The loud pattern at tide 8 produces an alarm squad within 90 s in a single region; the spread pattern produces none in the same time.

- ☐ Killing every cleaner in a region raises host aggression map-wide within 5 s and it decays after 300 s.
- ☐ A downed diver (8 blood/s) draws the region's tier-3 patroller within 45 s.
- ☐ The camping pattern loses to the ecosystem (bot team dead) by tide 14 ± 3 ; the quiet pattern reaches tide 22 ± 4 .

The Sunken Ship

- ☐ 15 sprats killed in the salon calls the barracuda within 30 s and the bull shark within 90 s.
- ☐ Blood made in the salon reads at the stern within 40 s (current).
- ☐ Feeding the crocodile a turtle suppresses its attacks for 60 s.
- ☐ The Goliath inhales from 4 m and releases at 150 gill damage.

The Underwater Cave

- ☐ A wounded eel entering slipstream A is smelled in the Chimney 5 s later and the Sump 11 s later.
- ☐ A Drowned squad walked past the Ghost Worm's wall loses two Deckhands.
- ☐ The sturgeon's death (80 blood) brings the sleeper shark within 90 s.
- ☐ Rockfall triggers at 5% per noise-6 shot in the Chimney and never elsewhere.

The Coral Reef

- ☐ Crown-of-thorns erode one corridor segment per 90 s and a shark uses the gap within 2 minutes.
- ☐ The dolphin pod rams a shark within 10 m of a diver in under 3 s; one dolphin hit by a diver flips the pod for the match.
- ☐ The tide reverses every 4 minutes and blood on the Flats reaches the Wall in 40 s on the ebb.
- ☐ The third drum beat calls a shark; the fifth calls the Matriarch.

Atlantis

- ☐ The generator places 60+ buildings by district with no overlaps and every catalogue count met.
- ☐ Plaza blood reaches the market in 75 s and the outfall in 120 s.

- ☐ Dead Lost Ones repel the sandbar shark within 20 m for 60 s and draw crabs within 30 s.
- ☐ Netting three grates in a district forces the Wyrms to the fourth.

Approaching the Void

- ☐ Salamanders and coelacanths reach their caps (12) within 20 minutes if uncontrolled; the caiman alone holds them at about 6.
- ☐ A whale kill at Overlook 2 brings the Relict within 3 minutes.
- ☐ Crossing the bone stakes consumes a diver after 8 s of tremor, every time.
- ☐ The Leviathan's Dark puts out every light within 30 m for 6 s and its lure remains.

Engine and systems workbook

`RedTide_Engine_and_Systems.xlsx` holds the numbers that are not per-map. Claude Code loads it as `redtide_engine/*.json`.

Sheet	Holds
Constants	80 tuning constants for scent, blood, senses, sound, alarm, hunger, states, social behavior, population, symbiosis, flora, enemies, and the player, each with its intended effect and a worked check
Movement	The diver's body: swim, sprint, stamina, vertical, acceleration, aiming speed, walking in air, grab-break rules, knockback and stun defaults, interaction holds, the void's pull
Drops	Drop weights by tide era, the anti-repeat rule, and the Locker pool weights with the eel's move rule
Progression	Token sources, cosmetic costs, and the 50 ranks with their unlocks
Dossier_ship, _cave, _reef, _atlantis, _void	A generated dossier page for every species and flora: class, size, tier, diet and predators (derived from each map's diet matrix), weak point, field note, and how to handle it

Sheet	Holds
Barks	252 diver lines across 21 situations (3 per diver per situation) plus the voice rules for expanding each set to 6–10

Movement, in one paragraph

Divers swim at 2.0 m/s, sprint at 3.4 for 6 s on a stamina bar that only appears while sprinting and refills in 8 s, rise and dive at 1.4, and reach full speed in 0.4 s. Aiming slows swimming to 60%; harpoons and spearguns to 40%. In air chambers they walk at 1.6 with no jump and a 1 m step-up. A hold can be broken early by mashing (six taps) or by a teammate dealing 15% of the holder's HP. Knockbacks unsettle aim for 0.6 s. Purchases over 1,000 need a 1 s hold; reviving requires holding still; downed divers drift-crawl at 0.6. Past the abyss edge the pull lasts 5 s with a warning at the first second.

Progression, in one paragraph

A match pays 10 tokens, a win (tide 10+) 25, and 2 per tide past 10; firsts pay more (tide 20 on a map 50, a boss 30, a quest 100, a full dossier 75). Ranks cost $50 \times \text{rank}^{1.6}$ tokens each, so a good match earns 40–70 and rank 50 is about 150 matches. Ranks unlock charms, skins, pouch slots, and three bonus dossier pages; two charms (Lucky Locker, Fair Shares) come from tide milestones instead. Cosmetics cost 100–300. Nothing bought or earned changes in-match power.

The how-to-play page (final text)

One scrolling page in the arcade menu, styled as a page torn from the ship's log, chalk headings, and one diagram. This is the game's only teaching; nothing in a match explains itself.

THE OWNERS' STANDING ORDERS FOR DIVERS

1. The job. You and up to three others go down with a pistol, a knife, two charges, and five hundred scrip. You kill what lives there and the Owners pay by the head. Every tide sets a quota; meet it and the next tide is worth more and wants more of you. There is no end. There is only how deep you get.

2. Scrip. Ten for a hit, the bounty for a kill, more for the eyes and the gills. Scrip opens doors, buys guns off the walls, pulls from Davy's Locker, pours tonics, and feeds the Pressure Forge. It's yours alone; the Fair Shares charm is the only way to split it. Spend between tides. Twenty seconds of calm is all you get.

3. Blood. Everything you shoot bleeds, and everything you kill bleeds more. Blood drifts with the current and the water remembers it. Small things smell a little blood and come to eat. Big things need a lot of blood, but ten small kills make as much as one big one. Scavengers clean it up; kill them and nothing does. Watch the red at the edge of your mask, watch the vial, and when the pulse starts, leave.

4. Noise. Guns are heard. Fire enough in one place and someone comes to see who's spending the reef: Wreckers, the Drowned, the tribe, the Lost Ones, what's left of the station. They hunt you, and the reef hunts them, and the smart diver stands where the second thing happens. Four divers firing in one room bring them four times faster than four rooms. Knives bring nobody.

5. Going down. At zero you're floating with the pistol and forty-five seconds, and you're bleeding into the water the whole time. A friend can pull you up in four; fewer with Quick Brine. Dead is when everyone's down. Cold Coffin and Quick Brine buy second chances; nothing buys a third.

6. The rest you learn by dying. Which coral grabs. Which light isn't a light. Which fish is on your side until you shoot it. Which door the shark can't fit through. The dossier fills in as you go: every beast you kill or watch for half a minute tells you what it eats, what eats it, where it's soft, and what it's afraid of. Read it between matches. It's the only map that matters.

Diagram caption (a chalk drawing of a diver, a dead fish, and three arrows): *One kill: a little blood, the scavengers come. Ten kills in one place: the shark comes. A diver down: everything comes. Move, spread out, keep it quiet, and let the water forget you.*

Below the page, in smaller chalk: Controls are on the back. Good luck. The Owners are not responsible for anything that happens below the surface, including the surface.